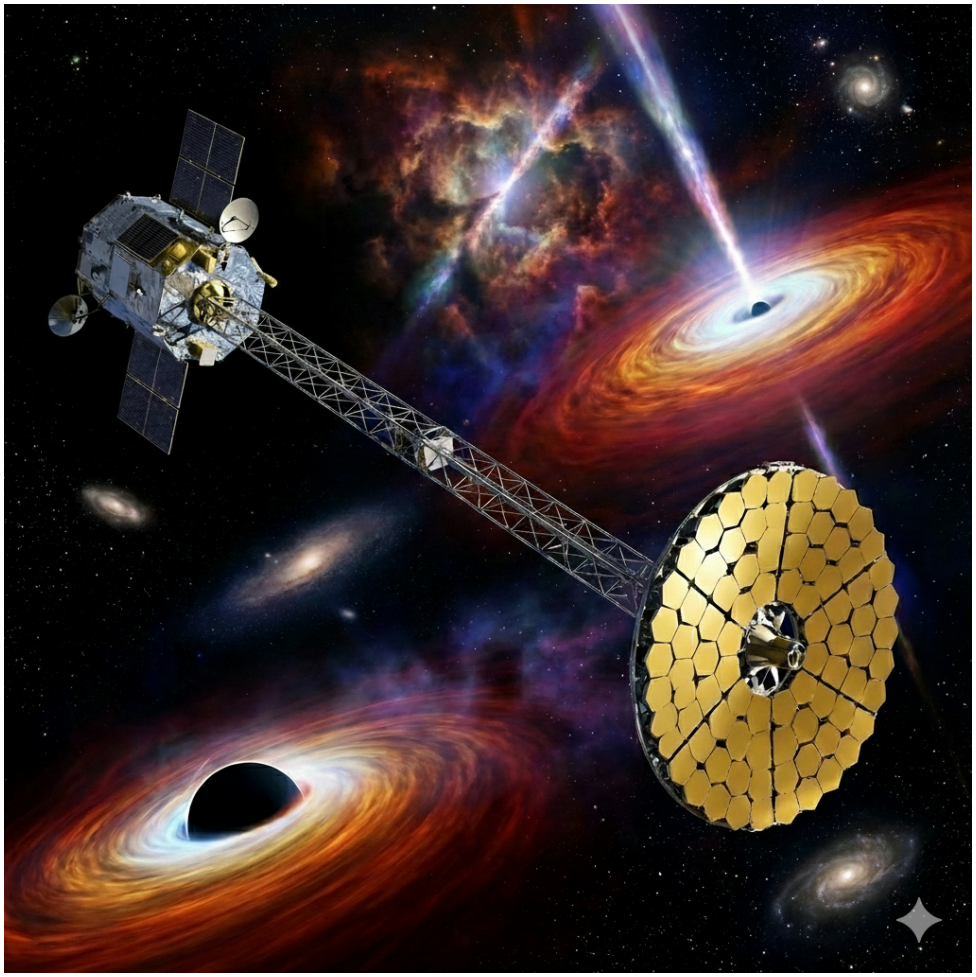


PHEMTO: The Polarimetric High Energy Modular Telescope Observatory

Proposal for the M8 ESA Call

"Exploring the violent and magnetic Universe via hard X-ray spectro-imaging polarimetry"

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Executive Summary

With the opening of the X–ray and gamma–ray windows in the sixties, extremely energetic and violent phenomena were discovered and subsequently found to be ubiquitous in the Universe. The existence of compact objects, in particular black holes, was unveiled. In addition to their obvious interest as objects predicted by General Relativity, it has also been realized that black holes of all masses have a fundamental role in shaping the observable properties of the Universe.

Observations in the high-energy domain are fundamental for understanding how matter is organized and behaves around black holes; unravelling how these extreme objects influence their environments on a very large scale; and finding the still elusive obscured massive objects in the center of galaxies. In addition, polarimetry in the hard X-ray domain is an important and unique new diagnostic tool at energies for which the non-thermal processes dominate. Acceleration phenomena, non-thermal processes, and magnetic field topology of all types of sources can be probed using this new and independent observable.

Other major problems in contemporary astrophysics, such as the understanding of acceleration processes at shocks of all sizes in relation to the origin of cosmic rays, or the definitive characterization of the debated non-thermal X-ray energy content of clusters of galaxies, also require observations at high energies. An observatory-type medium mission operating from 1 to 400 keV with excellent characteristics in terms of imaging, spectroscopy, timing, and polarimetry, such as PHEMTO, can provide direct insights into these major fields of research. **Its three orders of magnitude improvement in sensitivity in the hard X-ray band will be a real breakthrough and enable the observation of a wide diversity of astrophysical sources unreachable before.**

Table 1: Summary of *PHEMTO* characteristics

Key Science Goals	• Understand the role of magnetic field in cosmic accelerators • Probe the nature of accretion/ejection in accreting black hole and neutron stars systems • Constrain explosion physics in supernovae and their remnants • Describe the structure and demography of obscured AGN • Determine particle acceleration and magnetic fields in clusters of galaxies
Payload	Soft/Hard X-ray imaging polarimeter with a 20 m focal length (EOB) composed of: • X-ray mirror, $1-80$ keV, FOV ~ 10 arcmin, angular resolution ~ 10" (HPD) • Hard X-ray Laue Lens, $50-400$ keV, FOV ~ 5 arcmin, angular resolution ~ 25" (HPD) • Si/CdTe Compton telescope with active/passive shielding
Mission Profile	• Ariane 6.2 • Duration: 5-year (nominal) + 3 years (extended) • Orbit: LEO, $< 5^\circ$ inclination
Spacecraft	• 3-axis stabilized • Power: 500 W • Dry Mass: 1400 kg • S-band/X-band (Malindi, Kourou)

1. Science Cases

With its unique dual optical system providing high sensitivity over more than two decades in energy ($\sim 1-400$ keV), PHEMTO will answer two of the main themes describing the Hot and Energetic Universe, one of the fundamental topics of the Voyage 2050 ESA roadmap. These themes are the study of the strong magnetic fields in our Galaxy from compact objects to supernovae and the study of particle acceleration from black holes (of all sizes) to supernovae as well as in clusters of galaxies.

These two themes will be also in very strong synergy with the observatories roadmap foreseen at other wavelengths, such as SKA in the radio domain, the Vera Rubin telescope in optical, NewAthena in X-rays and CTAO at very high gamma-ray energies. We have divided these scientific themes into five questions of utmost interest for high-energy astrophysics:

- What is the role of magnetic fields in our Galaxy?
- How accretion links to ejection in accreting black holes of all masses and neutron star systems?
- How to constrain explosion physics in supernovae and their remnants?
- What is the demography and role of supermassive black holes through cosmic time?
- How are particles accelerated in galaxy clusters?

1.1 The Role of Magnetic Field in Cosmic Accelerators

Cosmic rays (CRs) represent a fundamental component of the interstellar medium, contributing comparably to magnetic fields, turbulence, and thermal gas in the overall energy budget of galaxies, while playing a critical role in driving galactic winds, regulating star formation, and in chemical evolution via ionization processes. Despite their profound impact on galactic ecosystems, the precise origins and acceleration mechanisms of CRs remain among the most compelling open questions in high-energy astrophysics.

At the galaxy scale, the fast shocks of young SuperNova Remnants (SNRs) are one of the main efficient particle accelerators, as evidenced by X- and gamma-ray observations of these targets [Vink, 2012, Lemoine, 2025]. Another class of interesting accelerators are Pulsar Wind Nebulae (PWNe), bubbles of relativistic particles (mostly electron-positron pairs) inflated by pulsar winds that form a relativistic shock which accelerates particles up to PeV energies [HESS Collaboration et al., 2016]. These sources have recently drawn some attention as potential counterparts of the ultra-high energy gamma-ray ($E > 100$ TeV) sources detected by the water Cherenkov telescope LHAASO [Breuhaus et al., 2022, Cao et al., 2024]. At a much larger scale outside of our Galaxy, Active Galactic Nuclei (AGNs) sometimes show collimated, relativistic jets that are thought to be cosmic accelerators too, capable of propelling particles to extreme energies on scales ranging from parsecs to kiloparsecs. Although the basic concepts of diffusive shock acceleration are understood, many important aspects remain unexplored, such as the maximum energy reached, the amplification of seed magnetic fields, the level of magnetic turbulence, and the magnetic configuration that can be tested at a variety of scales from km to kpc (from neutron stars to AGN jets). Because polarized X-rays are emitted by the highest energy electrons closest to the acceleration sites and are less affected by depolarization than in radio and optical, high angular resolution hard X-ray imaging and polarimetry is the ideal way to study the acceleration processes and measure the properties of the magnetic field in particle accelerators.

The Role of the Magnetic Field in SNRs and PWNe

- **Particle acceleration and turbulence via the study of the synchrotron emission and polarization.** Observations of synchrotron X-rays prove that young supernova remnants accelerate electrons, perhaps up to energies of 100 TeV or even higher. Yet the maximum energies these particles can reach and the details of the acceleration mechanism are still unclear. For most young SNRs, the synchrotron spectral cut-off is in the X-ray band and its shape can provide important constraints on the acceleration mechanism,

particularly on the diffusion coefficient α ($D \propto E^{\alpha}$) and turbulence regime [see e.g. Yamazaki et al., 2014]. Fig. 1.1 shows the example of the young SNR Cassiopeia A where the synchrotron cut-off can be fully covered in the PHEMTO energy band.

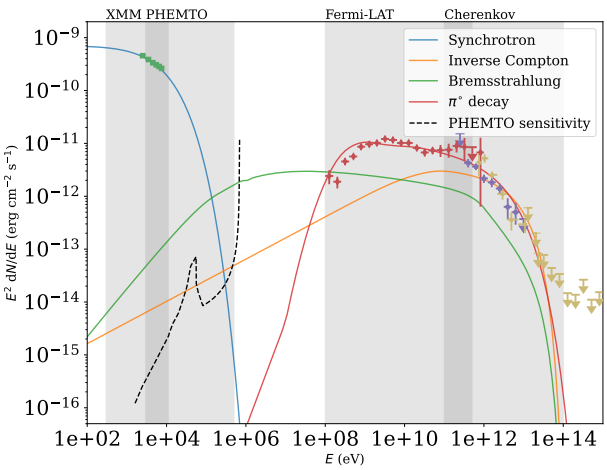


Fig. 1.1: Spectral energy distribution showing the different processes at play in SNRs exemplified with Cassiopeia A. Details of the synchrotron curvature, the inverse Compton, and the non-thermal Bremsstrahlung emission will be detectable in a 100 ks PHEMTO observation.

We note that what we observe is the synchrotron radiation and not the particles' properties directly. The link between the electron population and the observed photons spectral signature depends on magnetic field configuration (uniform or turbulent) and needs to be constrained simultaneously by polarization measurements. NuSTAR observations have allowed us to probe the high-energy spectrum of the Cassiopeia A SNR (the brightest X-ray SNR) but only up to ~ 50 keV. The SNR maps remain completely unexplored by focusing imaging instruments in the $50-400$ keV band and PHEMTO will provide the first spatially resolved images above 50 keV. This can be used to study the inverse Compton γ-ray emission in SNRs to probe the properties of the electron population and estimate the magnetic field strength by comparison with the X-ray synchrotron emission. In the gamma-ray regime, such study is currently limited due to: 1) the emission being a mixture of electron and hadron emissions (via the π^0 process, see Fig. 1.1), scrambling the link between the photon spectral properties and the electron properties, and 2) the spectral properties of the electron population and the estimate of the magnetic field strength can only be obtained at the level of the entire SNR, due to the present hard X-ray telescopes limited angular resolution. **The 1 – 400 keV PHEMTO energy band, combined with high angular resolution has the unique capability to resolve and measure the full spectral properties of the electron population.**

Concerning polarization, using a pixel-by-pixel analysis, IXPE was not able to map the faint polarization signal [Vink et al., 2022]. Spatially resolving and mapping the shock structure at the 5 – 10" scale both spectrally and in polarization is required to fully understand the acceleration key questions, and was out of reach for IXPE 25" spatial resolution. Another open question that is within reach of PHEMTO's higher-resolution and sensitivity is to study how the magnetic field flips from tangential (due to shock compression) to radial (due to instabilities) within that narrow shock region [Vink et al., 2022].

With the PHEMTO sensitivity and 10" angular resolution, it will be possible to map the shock rim in CasA and several other SNRs (e.g. Tycho, Kepler, SN 1006, Vela Jr, RX J1713-3946) spatially, spectrally, and in polarization, opening the door to population studies.

- **Magnetic configuration in pulsar wind nebulae.** X-ray images of many young PWNe reveal axisymmetric features with a torus and jet, like in the Vela and Crab nebulae. The torus can be explained by an anisotropic pulsar wind energy flux (higher in the equatorial plane) interacting with a toroidal magnetic field which is compressed and accelerates particles. However, to produce jets, an additional magnetic component is necessary, either a poloidal or disordered (isotropic) field. The observations by IXPE of the Crab nebula have revealed that the polarization is not uniform across the nebula and shows strong spatial variations. For example, in the Crab while the integrated polarization degree is ~ 20 %, very localized regions can reach values as high as 50 % of polarization degree [Bucciantini et al., 2023]. Interestingly these highly polarized regions are located at the edges of the torus and do not correspond to the brightest X-ray features. In addition, in both the Crab and MSH 15-52, the base of the jet shows

low polarization [Romani et al., 2023, Wong et al., 2024]. It is however unclear if this is due to intrinsic magnetic reconnection and turbulence or simply contamination from the brighter torus that overlaps at IXPE's resolution [Wong et al., 2024, Di Lalla et al., 2025]. PHEMTO increased sensitivity and better angular resolution than IXPE will allow to better resolve the inner structures of the PWN (pulsar, torus, jets, wisps, and knots) and therefore reduce the leakage of bright but low polarized region towards fainter but higher polarized regions. **Only resolved PHEMTO X-ray polarization maps will allow to study PWN systems at high angular resolutions and to spatially resolve the polarization properties of its individual spatial structures** [Bucciantini et al., 2017].

Magnetars as Powerful Cosmic Accelerators

Magnetars are isolated X-ray pulsars whose emission is thought to be powered by the dissipation and decay of their ultra-strong magnetic fields ($B \sim 10^{13}$–10^{15} G), which are several orders of magnitude larger than those of regular rotation-powered radio pulsars. They were initially classified in two groups: Anomalous X-ray Pulsars (AXPs), exhibiting X-ray luminosities higher than their rotational energy loss rate, and Soft-Gamma Repeaters (SGRs), characterized by the repeated emission of energetic short bursts. Nowadays it is clear that both SGRs and AXPs host magnetars, which can show a continuous spectrum of behaviors, from long periods in quiescence to strong bursting activity. Indeed, the magnetar hallmark is their unpredictable flaring activities observed in the X-rays and gamma-rays on different time scales, ranging from milliseconds to minutes (i.e., bursts and giant flares), and longer-lived X-ray outbursts (from months to years). These sources are believed to play a crucial role in a much wider range of astrophysical phenomena, from ultra-luminous X-ray sources (ULXs) and fast radio bursts (FRBs) to (cosmological) gamma-ray bursts (GRBs). Moreover, magnetar-like activity has been discovered in other types of isolated neutron stars (NSs), such as two high-magnetic-field radio pulsars and one central compact object.

- **Physics of the magnetar high-energy emission.** Many magnetars exhibit high-energy hard tails that extend up to several hundreds of keV, either persistently or following transient events, with luminosity generally comparable with or larger than that measured below 10 keV. First observed with INTEGRAL from persistently bright sources and later confirmed by Suzaku and NuSTAR, the hard X-ray component is adequately described by a power-law-like spectrum with photon index $\sim 0.5-2$. Upper limits in the hundreds of keV and MeV regions suggest that these tails do not extend above ~ 500 keV. A peculiarity of the hard X-ray components is that the spectrum of the pulsed emission is harder than that of the averaged hard X-ray one and also shows phase dependent variations. Moreover, they vary in time when the source is undergoing an outburst: the non-thermal tail is detected at the outburst peak and is observed to fade on faster time scales than the softening of the 0.3 – 10 keV X-ray spectrum.

None of the current missions offer the combination of sensitivity and broadband energy coverage needed to study them in depth. As a result, the nature of this emission remains unclear in terms of radiation and acceleration processes. Multiple mechanisms at their origin have been proposed: bremsstrahlung photons emitted by the neutron star surface layers heated by returning currents, synchrotron emission from pairs produced at a height of about 100 km above the neutron star, and resonant Compton up-scattering of soft thermal photons by a population of highly relativistic electrons threaded in a twisted magnetosphere or in a confined bundle. These scenarios manifest themselves with different polarization signatures. IXPE observed five magnetars that show persistent emission at hard X-rays. However, only in one case, for the magnetar 1E 1841–045 [Rigoselli et al., 2025], it was possible to derive polarization measurements for the hard X-ray component. Thanks to a simultaneous NuSTAR observation that ensured a correct spectral decomposition over a broad energy range (up to ~ 80 keV), the contribution of the hard non-thermal tail was identified to dominate above ~ 5 keV and a spectro-polarimetric analysis found it polarized at more than 65 %, pointing to a synchrotron/curvature origin. For the other sources, this component was out of reach of the IXPE band. **With its broad energy range coverage, PHEMTO will provide the key to unambiguously determine the presence of the hard tail and its polarimetric properties, and**

in assessing their origin. Moreover, a sensitive spectro-polarimeter like PHEMTO will reveal the magnetic field topology, locate the emission region(s) in the magnetosphere, map for the very first time the particle acceleration along closed field lines and quantify its attenuation due to photon splitting and pair production in high B fields.

High-energy emission is also detected during short and intermediate bursts ($\sim 10^{-2} - 50$ s, energy released $E \sim 10^{36} - 10^{43}$ erg) and giant flares ($\sim 10^2 - 10^3$ s, $E \sim 10^{44} - 10^{47}$ erg). Currently, a fully coherent model capable of explaining what triggers these events is still lacking. The most promising mechanism suggests that magnetar bursting activity arises in a hot electron-positron plasma that remains trapped within the closed magnetic field lines, the so-called trapped fireball model [Thompson and Duncan, 1995]. Simulations showed that the linear polarization degree is expected to be high in the PHEMTO energy band, varying between ~ 70 % and ~ 50 % depending on the viewing geometries, while the polarization angle is predicted to be constant [Taverna and Turolla, 2017]. **Thanks to its sensitivity, PHEMTO will easily measure these quantities, helping us to test this emission mechanism for magnetar flares**.

It is also worth mentioning the detection of a short non-thermal X-ray burst from the magnetar SGR J1935+2154 temporally coincident with a bright radio burst, whose properties were reminiscent of those of FRBs, for the first time in April 2020 [see e.g., Bochenek et al., 2020, Mereghetti et al., 2020]. This discovery strongly supports models based on magnetar engines for extragalactic FRBs. Observed also with INTEGRAL in the 20 – 200 keV band, the X-ray burst stood out from the other bursts emitted from this source because of its harder spectrum with a high cut-off energy (~ 65 keV). This measurement is well within the capabilities of PHEMTO, despite the short duration of these events. Moreover, **PHEMTO will provide polarization data that, combined with radio data, will enable us to understand whether the X-ray and radio emission originate from the same region or different layers of the magnetosphere.**

- **System inclination and gravitational wave (GW) emission.** The geometry of a magnetar, specifically the inclination between its spin axis, magnetic axis and the line of sight, plays a central role in both its high-energy emission and its potential as a GW source. Models of non-thermal magnetar radiation show that the predicted spectrum and pulse morphology depend strongly on this geometry, yet current X-ray data focusing only on spectroscopy and/or timing studies cannot uniquely establish it. **Hard-X-ray polarimetry, as enabled by PHEMTO, offers the first direct route to breaking these degeneracies**: the polarization degree and angle encode the orientation of the magnetic field and the observer viewing angle, allowing reconstruction of the system configuration. Indeed, **by measuring polarization as a function of energy and NS rotational phase, PHEMTO can reconstruct the inclination between the spin axis, magnetic axis, and line of sight with unprecedented precision**. These measurements also provide a unique handle on the spin-velocity alignment imprinted by the natal kick, a quantity that determines whether the magnetar is an aligned or orthogonal rotator. Since orthogonal configurations maximize magnetic deformation and therefore GW emission, **PHEMTO polarimetric constraints become a gateway to quantifying the GW contribution of both newborn and mature magnetars**.

AGN jets as particle accelerators

Blazars are the most powerful and variable objects among the radio loud AGN. This is because strong relativistic aberration effects, primarily light magnification and time intervals foreshortening, take place in the plasma flowing at speeds close to that of light in their powerful jets, that are oriented only a few degrees away from the line of sight. Multi-wavelength studies of blazars in the last 25 years have identified a characteristic broad-band spectral shape, with two peaks in a ν-Fν (where Fν is the flux density at the frequency ν) representation. The observed synchrotron emission (first peak), from radio waves to X-rays, confirms the presence of ultra-relativistic electrons, while the higher-energy components (second peak) suggest efficient acceleration processes linked to internal shocks, magnetic reconnection, or shear structures in the flow [Lyutikov, 2003, Hervet et al., 2017]. In this context, X-ray and gamma-ray

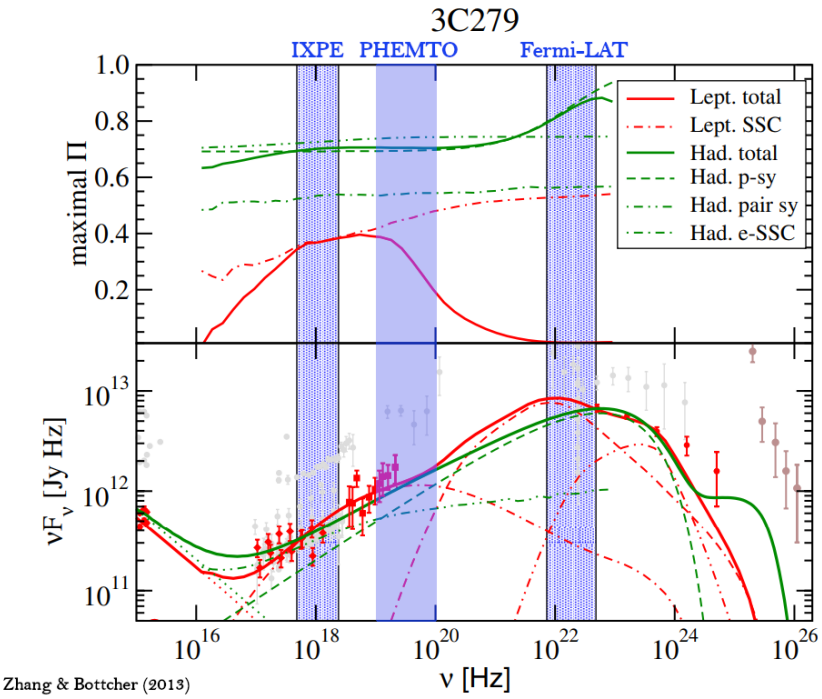


Fig. 1.2: Ultraviolet through γ-ray spectral energy distribution (lower panel) and maximum degree of polarization (upper panel) for the blazar 3C 279. Leptonic model fits are plotted in red and hadronic models in green. Different lines indicate individual radiation components as labelled in the legend. Shaded areas indicate the 2 – 8 keV IXPE, 40 – 400 keV PHEMTO and 30 – 200 MeV Fermi-LAT range in which high energy emission may be measurable. Figure adapted from Zhang and Böttcher [2013].

polarimetry offers a particularly powerful diagnostic tool: it allows us to constrain the composition of the jets (leptonic versus hadronic), measure the order of the polarization degree and the geometry of the magnetic field, and locate the dominant regions of particle acceleration, as shown in Fig. 1.2. In particular, a high degree of polarization and consistent variations in the polarization angle favour globally ordered magnetic fields or reconnection-dominated scenarios, while weaker or chaotic polarization points to regions dominated by turbulence or multiple shocks. In low-synchrotron-peaked blazars, hadronic models predict substantially higher maximum degrees of X-ray and γ-ray polarization than leptonic models. In contrast, for high-synchrotron-peaked blazars, where the X-ray emission is dominated by electron synchrotron, both leptonic and hadronic scenarios yield similar X-ray polarization levels, while hadronic models still predict significantly higher maximum γ-ray polarization. **PHEMTO is therefore ideally positioned to discriminate between these scenarios and to probe both the jet composition and magnetic field topology in these cosmic accelerators.**

1.2 Accretion and Ejection Physics

Accretion is the most efficient way to produce energy. It powers a wide range of astrophysical objects at all ages of the Universe, ranging from distant gamma-ray bursts to close-by planet forming systems. The most violent and rapid manifestations of accretion phenomena are associated with black holes and neutron stars. More than fifty years of multi-wavelength observations of accreting objects have shown that relativistic ejections are seemingly associated with specific phases of accretion through mechanisms that are yet to be understood. In addition, studies of accretion in highly magnetized, young neutron star enable to probe the behaviour of matter in extreme magnetic fields.

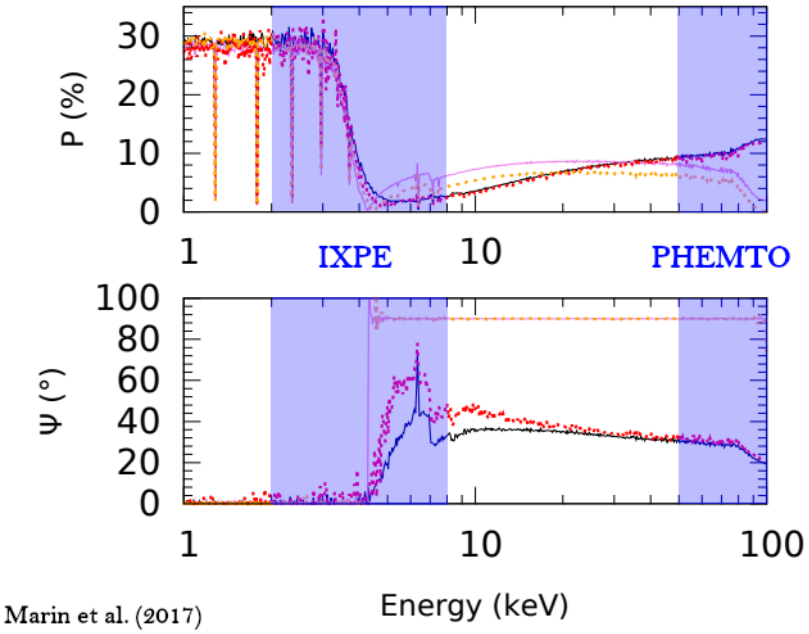


Fig. 1.3: Polarization degree (top) and polarization angle (bottom) of a type-2 (edge-on) radio-quiet AGN simulation obtained with the STOKES code [Marin et al., 2012] for four different polarizations of the X-ray corona, representing different energies and compositions. While the IXPE (2 – 8 keV) band suffers from selective absorption and dilution by emission lines, the PHEMTO band is clean from any polluting features and can determine the polarization state of the corona photon with higher accuracy.

- **Accretion and ejection physics in black holes at all mass scales.** In AGN and X-ray binaries (XRB), the bulk of the emission is released in the X-ray and soft gamma-ray bands, i.e, typically in the range of ~ 1 keV to a few MeV. **The spectral window of PHEMTO is thus ideal to probe the physics of accretion**. It has, however, become clear over the past decades that the diagnostics enabled by spectroscopy alone are not sufficient to model theoretically the broad band emission in accreting sources [Nowak et al., 2011]. In addition to timing studies (an advent of the 90's – 2000's era), polarimetry provides a new, independent and complementary window onto this physics, providing access to new parameters such as the system's inclination and geometry or magnetic fields strength.

The currently accepted contributors to the high-energy emission from stellar and supermassive black holes and neutron stars with a low magnetic field, are an accretion disk, a Comptonised medium (referred to as the 'corona', whose origin, geometry and global properties are still badly constrained), and possibly a jet. These media contribute to the overall emission in different proportions. This already rather complex picture has been further complicated by the discovery of disc winds in disc dominated states, seen through absorption lines below $\sim$ 10 keV [e.g. Ueda et al., 1998, Neilsen and Lee, 2009], and hard tails (100 keV – 10 MeV) in various states [Grove et al., 1998, McConnell et al., 2000, Laurent et al., 2011, Jourdain et al., 2012, Cangemi et al., 2019]. The launching mechanism of the winds and the origin of the hard tails are highly debated.

The evolving, relative contribution of all these components permit to distinguish various emission states related to very different flavours of accretion/ejection properties, and thus different physical conditions in the vicinity of the compact object. How do these media interact? What are their exact geometry and origin? What triggers their spectral state evolution and transition? What are the intimate links between accretion and the ejection of matter at relativistic speeds? What is the specific influence of the very large gravitational field, that of the extreme magnetic fields? These are amongst the most important questions we seek to answer.

X-ray polarimetry, as already demonstrated by IXPE, is a unique tool to study accretion into a deep gravitational well and measure relativistic aberration and beaming, gravitational lensing, and gravito-magnetic frame-dragging. In particular, X-ray polarimetry can probe the geometry and location of the corona [Gianolli et al., 2023], but also the reprocessing of coronal photons onto the accretion structure at higher energies, where the accretion disk behaves like a mirror, polarizing the scattered radiation [Marin et al., 2012]. The advantage of hard X-rays is that they are less absorbed by the cold, circumnuclear reservoir of gas and dust that surrounds AGNs, and thus carry information from the central engine at any inclination angle. As shown in Fig.1.3, **PHEMTO will thus be able to distinguish the relative contribution of the X-ray corona and the disk-reprocessed component, while determining the overall geometry of the system at the end of the Compton hump.**

The innovative potential of X-ray/soft gamma-ray polarimetry has also been glimpsed by the INTEGRAL polarimetric studies of various sources such as the Crab pulsar/nebula, GRBs and a few bright XRBs showing highly polarized emission above 400 keV [Götz et al., 2013, Rodriguez et al., 2015, Cangemi et al., 2023, Bouchet et al., 2024], probably associated with synchrotron emission from the jet. The tension between the polarized properties at soft X-rays and those measured above a few hundred keV is however puzzling. A tentative theoretical explanation was put forth by Bouchet et al. (2025, PhD thesis), and provide strong predictions for the polarization properties in the 10 – 100 keV range.

By bridging the spectral gap between IXPE and INTEGRAL (and improving the sensitivity of the latter), PHEMTO will disentangle different polarized emission from these sources in different states. **PHEMTO will open the 50 – 400 keV range to high sensitivity spectro-polarimetric studies of accretion/ejection for the first time.**

- **Accretion in highly magnetized neutron stars.** An important subclass of X-ray binaries comprises a high mass star and a neutron star as accretor: these are known as *neutron-star high-mass X-ray binaries* (NS-HMXB). Since the neutron star is relatively young, it is generally endowed with a strong magnetic field of the order of 10^{12} G which might imprint characteristic signatures in the hard X-ray spectrum, known as cyclotron line scattering features [see Staubert et al., 2019, for a review].

These features appear in absorption over a smooth exponentially decaying continuum with a width of a few keV at an energy between 10 and 120 keV, and they constitute the only direct measurement of the neutron star's magnetic field in its immediate vicinity. Despite this phenomenon being observed since 1976 on a few tens of sources, a lively debate is still present on the detailed physical mechanisms that control the appearance of these features, owing to the inherently complex physics in place.

When radiation propagates in a highly magnetized medium, it gets polarized both in the scattering process and in vacuum. There are two modes of polarization known as ordinary and extraordinary depending on the location of the line-forming region and on the actual geometry of the accretion flow producing the X-ray emission. For instance, it is possible that cyclotron scattering features are formed at the boundary of the magnetically confined accretion column, or on the neutron star surface [Becker et al., 2012, Poutanen et al., 2013]. The relative contribution and localization of the accretion columns plays a crucial role, but this information is difficult to obtain. Techniques of phase-resolved spectroscopy and characterization of the amount of pulsed emission contain still a high degree of degeneracy.

Observations of several HMXB with IXPE revealed a degree of polarization in the 2 – 10 keV band of not more than about 15 % [Poutanen et al., 2024]. In this range, radiation was expected to be strongly polarized because of the strong dependence of the cross section on the kind of polarization [Caiazzo and Heyl, 2021a] and there is no consensus on the origin of this discrepancy. Application of the rotating vector model has revealed to be adequate to describe the spin-phase dependency of the polarization angle, despite the field close to the neutron star is thought to deviate from the dipole. However, the magnetic field is so intense that it locks the polarization to the field lines until its intensity diminishes under a certain threshold. The distance is believed to be such that only the dipolar component is still relevant. The polarization angle spin dependency can, therefore, be used to constrain the location of the magnetic poles

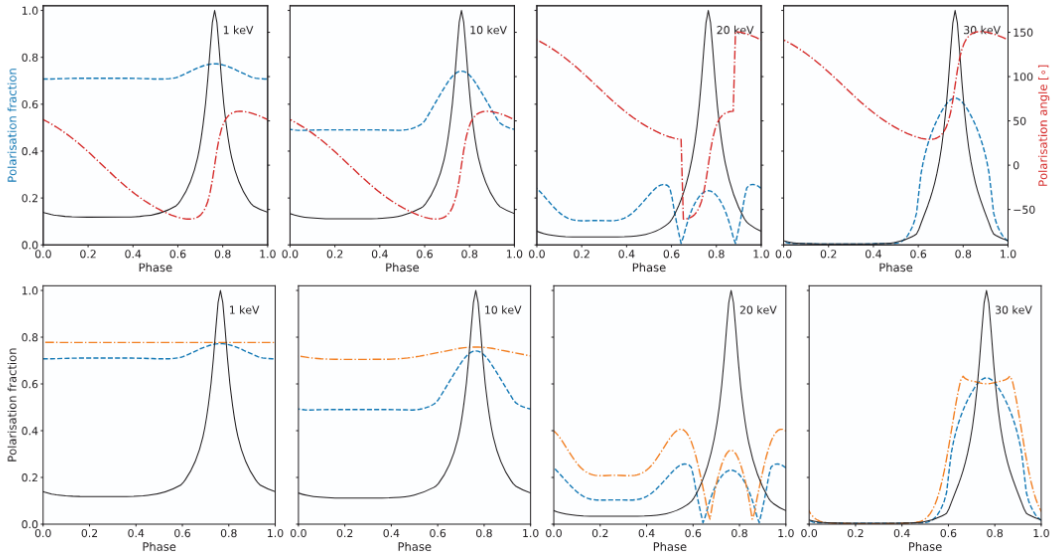


Fig. 1.4: The *upper panels* show the pulse profile (black), polarization fraction (blue dashed) and polarization angle in the base model of Caiazzo and Heyl [2021b] for Her X-1 (cyclotron line at ∼35 keV. The *lower panel* shows the effect of QED effects on polarization fraction in the orange dot-dashed line. [from Caiazzo and Heyl, 2021b].

on the surface, where the accretion stream impacts the stellar surface, module the uncertainty on which mode dominates the polarization.

It has been shown long ago [Meszaros et al., 1988] that the polarization properties of X-ray pulsar change dramatically when the photon energy crosses the cyclotron energy. This is intrinsic to the birefringence of plasma embedded in a magnetic field and has been confirmed also by Caiazzo and Heyl [2021a,b, see Fig. 1.4]. By comparing the polarization properties in this region, we can remove degeneracy in the rotating vector model and get a hint on the reason why the level of polarization at soft X-ray is relatively low. Meszaros et al. [1988] have also shown that the polarization pattern is different if the radiation is emitted at the top of the accretion column (pencil beam) or from the sides of the columns (fan beam). For such a study, it is thus essential to measure polarization at different spin phases of the neutron star: in a pencil beam configuration, the polarization angle has a discontinuity *at pulse maximum*, whereas in a fan-beam, this happens at pulse minimum.

Even more strikingly, polarization must be different for reflection models and models with lines originating in the accretion stream itself, but a quantitative estimation has not been published, yet. Analysis of polarization data benefit from the development of spectral physical models for the accretion column [e.g. Becker and Wolff, 2007, Farinelli et al., 2016] combined with the geometrical constraints derived from the study of polarization and pulse profiles [Poutanen et al., 2024]. **With its large spectral coverage and polarimetric capabilities, PHEMTO has the unique potential to explore the cyclotron scattering features forming region in great detail and thus unveil the behaviour of matter in extremely high magnetic and gravitational fields at unprecedented energy.** We have performed simulation of two bright transients (1A 0536+262 and V 0332+53) as observed during their last giant outbursts in 2021 and 2015: around the respective cyclotron energies, the minimum detectable polarization is compatible with the level detected at lower energy by IXPE for observation of 10 ks in phase averaged spectra and in 100 ks in phase-resolved spectra.

1.3 Constraining Explosion Physics in Supernovae and their Remnants

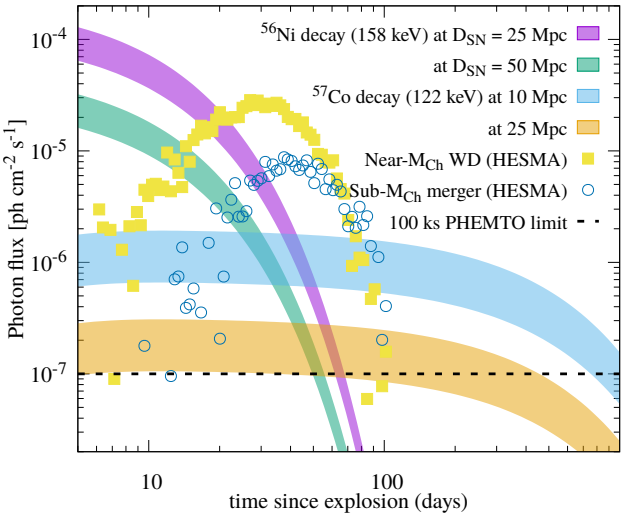


Fig. 1.5: Photon flux of decay lines of ^{56}Ni from type Ia SNe at various distances. The spread corresponds to a range of Ni mass of 0.3-0.6 $M_\odot$. Only radioactive decay is assumed without radiative transfer effects. Shown in blue and orange are the 122 keV line from the decay of ^{57}Co for a range of ^{57}Ni/^{56}Ni mass ratios from different explosion models, at 10 and 25 Mpc. Filled squares and open circles are output of angle-averaged HESMA simulations [Summa et al., 2013], accounting for optical depth and Compton scattering, for SD and DD scenarios, respectively, at 25 Mpc.

- Nuclear production in SNIa and progenitor channel. Runaway carbon burning in white dwarfs (WDs) approaching the Chandrasekhar mass M_{Ch} ($\sim 1.4\ M_\odot$) results in the complete nuclear incineration of the star, observed as a type Ia (or thermonuclear) supernova (SN). The optical light curve of supernovae is powered by the decay of radionuclides synthesized at the nuclear burning front. Type Ia SNe, in particular, are the main source of Fe in the Universe, from the radioactive decay of their main product ^{56}Ni to ^{56}Co and ^{56}Fe. This fact allows them to be widely used as standard candles to probe the high-redshift universe. Most notably, this provides strong evidence for the large amount and equation of state of dark energy (Ω_Λ and w parameters in cosmology [Riess et al., 1998, Perlmutter et al., 1999]).

However, when and how the explosions are triggered remain unclear, which introduces challenges to their use in cosmology [Miller, 2025] and chemical evolution models [e.g. Nissen et al., 2024]. Since WDs are inherently stable objects supported by the electron degeneracy pressure, interaction with another object must take place to trigger the thermonuclear runaway. Most of the established theoretical models involve the interaction with a companion star in a closed binary system. However, it is debated if the companion star is a second WD in the "double-degenerate" (DD) scenario [Iben and Tutukov, 1984, Webbink, 1984], or a non-degenerate star in the "single-degenerate" (SD) scenario [Whelan and Iben, 1973], and whether the explosion is triggered when the accreting WD approaches the Chandrasekhar limit or at an earlier stage. Understanding Type Ia supernova explosions therefore reduces to addressing two main questions: are the progenitors of Type Ia supernovae SD or DD systems? Is the explosion triggered in a Chandrasekhar-mass or sub-Chandrasekhar-mass WD ? Several observables can be used to point to the dominant progenitor channel, if any [Maoz et al., 2014, Ruiter and Seitenzahl, 2025]

The nuclear lines and high-energy continuum emission (60 – 400 keV, which is produced by the Compton scattering of line photons) provide the most direct measurement of the nickel mass synthesized in a SN Ia, $M(^{56}\mathrm{Ni})$, an essential constraint of the progenitor channel of SN Ia [Maeda et al., 2012]. At 0th order, the gamma-ray emission scales with $M(^{56}\mathrm{Ni})$. However, even if $M(^{56}\mathrm{Ni})$ is similar, various explosion models still have markedly different time evolution of line emission, especially for the 158 and 270 keV lines that fall within the PHEMTO lens bandwidth. As they are the lowest energy lines, they have the highest cross-section for Compton down-scattering and are therefore most sensitive to the velocity distribution of synthesized nickel and to the amount of surrounding material, both of which are different between merger and single-WD explosion scenarios [Summa et al., 2013].

Furthermore, the mass of another nickel isotope (^{57}Ni) can be observed from the 122 keV line of $^{57}\mathrm{Co} \rightarrow {}^{57}\mathrm{Fe}$ decay ($t_{1/2} = 272$ d). The ratio ^{57}Ni/^{56}Ni is sensitive to the physical conditions (central density and metallicity) of the exploding white dwarf [Leung and Nomoto, 2018], and is an efficient probe

of the higher neutronization occurring via electron-capture reactions in the dense core of near-M_{Ch} WDs, compared to sub-M_{Ch} WDs [Tiwari et al., 2022]

Direct detection of nuclear lines and continuum in the hard X-rays and gamma-ray bands and their early temporal evolution provide the best tools to measure $M(^{56}Ni)$, its velocity distribution, and the ^{57}Ni/^{56}Ni ratio, and thus address fundamental questions about SN Ia progenitors. Direct measurement using nuclear lines also alleviates uncertainties found in constraining Galactic SN Ia via chemical evolution models since the results depend on both the choice of core-collapse SN yields and the elemental ratio used as diagnostic [Trueman et al., 2025].

So far, only the nearest type Ia SN2014J has been detected via nuclear lines with INTEGRAL [Diehl et al., 2014, Churazov et al., 2014]. As the 3D nature of the explosion and viewing angle affect the line detection and measurement of Ni production [Summa et al., 2013], a larger sample of SN Ia is needed. For this, PHEMTO's three order of magnitude sensitivity improvement with respect to INTEGRAL/IBIS will dramatically increase the detection horizon: at an anticipated line sensitivity of 10^{-7} ph cm^{-2} s^{-1} for 100 ks, **the 158 keV and 270 keV ^{56}Ni $\rightarrow$^{56}Co decay lines can be observed routinely for SNe Ia within 50 Mpc (Fig. 1.5), or about one event per month.**

So far the ^{57}Ni/^{56}Ni ratio was measured in only 5 SN Ia [Tiwari et al., 2022], using very late-time bolometric light curve (owing to the slower decays of ^{57}Ni) requiring deep and challenging photometry. **Gamma-ray line detection provides a direct measurement of this ratio**. PHEMTO will be sensitive to 122 keV lines of SNe Ia within 25 Mpc (Fig. 1.5), allowing the first ever *direct* measurement of ^{57}Ni yield. This will allow to probe the innermost properties of exploding WDs in 1-2 SN Ia per year [Desai et al., 2026]. Coupled with high spectral resolution at softer X-rays from e.g. NewAthena/X-IFU to measure the yields of ^{55}Fe and other and neutron-rich species [Cr, Mn, Seitenzahl et al., 2015], we will be able to fully characterize the WDs that end up as type Ia SNe.

- **Probing explosions of stars with ^{44}Ti.** After short lifetime, the core of massive stars can implode and form neutron stars and stellar-mass black holes. In the process, they will most often produce a gravitationally powered supernova explosion called core-collapse (CC) SN. Asymmetries, electron fraction, and central temperature play key roles in the final outcome but require complex, multi-dimensional simulations [Burrows and Vartanyan, 2021]. Measurement of Fe-peak elements and Ti yields, products of the explosive silicon burning from the innermost regions of CC SNe provide strong observational constraints for these properties. In particular, ^{44}Ti is produced closest to the mass cut, the region separating material falling back on the newly-born neutron star from the ejecta. Its yield and the ratio ^{44}Ti/^{56}Ni are very sensitive to physical conditions [Boggs et al., 2015].

Since the decay of ^{44}Ti to ^{44}Sc (half-life of 59.2 yr) emits two lines at 68 and 78 keV, its yield can be measured in young supernova remnants (SNRs) up to a few hundred years old. The 3D distribution of this isotope can be revealed by resolving the emission both spatially (via imaging) and spectrally (via spectroscopy). Such measurements have been possible only for Cas A and SN1987A [Grefenstette et al., 2014, Boggs et al., 2015], but still suffer from large uncertainties.

Due to the jump in sensitivity by up to two orders of magnitude between NuSTAR and PHEMTO around 70 keV, we can reap the benefits ^{44}Ti can offer. It will be indeed possible to compare the 3D dynamics of that element with that of iron, the decay product of ^{56}Ni to constrain important aspects of the CC SN engine like the need for rapid rotation and/or jets [Wongwathanarat et al., 2017].

With the PHEMTO sensitivity the ^{44}Ti lines will be detectable for older Galactic CC SNRs such as Kes 75, 3C58, G21.5$-$0.9, and the Crab, for ^{44}Ti yields of $(0.5-2)\times10^{-4}$ $M_{\odot}$, thus offering new constraints on nucleosynthesis and explosion models [Chieffi and Limongi, 2017]. Since these remnants all contain pulsars or magnetars, the relative motion of titanium produced in the core of the star can be compared with that of the compact object to investigate the origin of neutron star kicks [Wongwathanarat et al., 2017]. Furthermore, stable ^{48}Ti is also produced in the innermost regions of SNe and can be observed as weak soft X-rays line, so further synergies with the NewAthena X-ray integral field units are expected, providing an essential probes of asymmetries in stellar explosions [Sato et al., 2021].

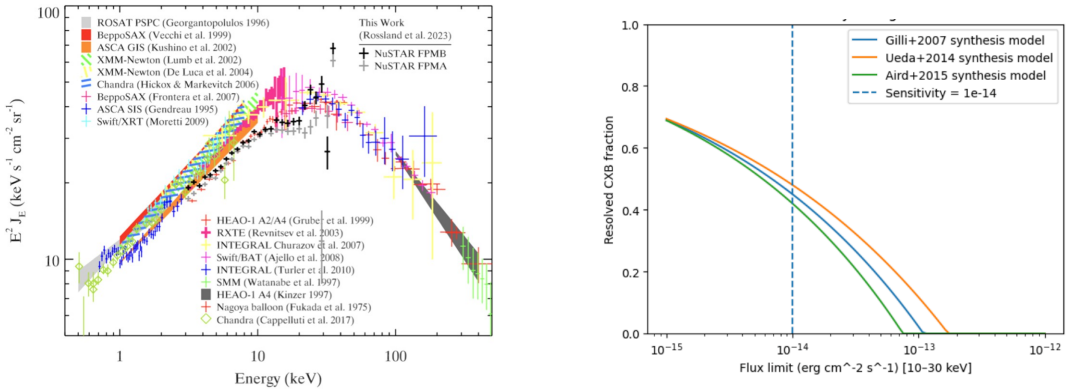


Fig. 1.6: *Left Panel:* Cosmic X-ray background measured by NuSTAR and previous missions (from Rossland et al. [2023]). *Right Panel:* comparison of the expected resolved CXB fraction in the 10-30 keV band, assuming current synthesis models.

In addition to remnants of core-collapse SNe, other Galactic remnants from type Ia SNe (such as G1.9−0.3, Kepler, Tycho) are close and young enough to allow resolving the distribution of ^{44}Ti, accurately measuring its yield, and comparing it to that of ^{56}Ni. This will provide a discriminating diagnostic between progenitor channels that was so far out of reach due to limited sensitivity [Tsygankov et al., 2016]. In particular, the ratio ^{44}Ti/^{56}Ni is several orders of magnitude higher for the detonated He shell than from the core [Shen et al., 2018]. Thus, mapping this ratio in type Ia SNRs would be the best direct test of the double-detonation model. So far this measurements was done only in 3 type Ia SNRs, and at a level sensitive to exclude only the double-detonation models that produce much higher ^{44}Ti/^{56}Ni ratio [Weinberger et al., 2020]. **PHEMTO will provide discriminating power between the more common SD and DD scenarios.**

1.4 Black Hole Census, Cosmic X-Ray Background and Obscured AGN

- **Cosmic X-ray background.** At hard X-ray energies ($>$ 10 keV), the CXB can be almost entirely traced to emission produced by accretion onto SMBHs. AGN of different types over a wide range of redshifts contribute to the CXB emission, tracing the accretion history of the Universe. For this reason, the spectral shape and overall flux of the CXB play a crucial role in constraining the various populations of AGN, especially that of the most difficult objects to detect, the Compton-thick (CT) AGN, i.e. those absorbed by a column density in excess of 10^{24} at/cm^2 and characterised by a strong reflected component peaking at around 30 keV. Since the CXB peak is located at around 30 keV, a large part of its emission comes from the most absorbed AGN (CT objects). Recently, NuSTAR has been able to account for $\sim$ 40% of the X-ray sources responsible for the CXB in the $\sim 8-24$ keV energy band [Rossland et al., 2023]. However, NuSTAR was not able to constrain the position of the CXB spectrum peak, confirming only that the peak lies above 20 keV. Figure 1.6 left panel, taken from [Rossland et al., 2023], shows the energy spectrum of the cosmic X-ray background as measured by all previous instruments from HEAO-1 to NuSTAR. As can be seen from the figure, NuSTAR measurements between $\sim$ 20 and 40 keV (i.e. the peak of the CXB spectrum) are poorly constrained: **this is the energy band where PHEMTO could add its own contribution towards solving this issue**. With an observing time of 1 Ms we will be able to reach a sensitivity of $\sim 10^{-14}$ erg cm^{-2} s^{-1} in the 10 - 30 keV band. From simulations (see Figure 1.6, right panel), we have computed that PHEMTO will resolve, depending on the synthesis model used, about 50% of the CXB compared to a resolved fraction of $\sim$ 5% in current Swift and INTEGRAL surveys. This fraction is comparable with that achieved by XMM and Chandra between 5 and 10 keV, and still larger than NuSTAR in this energy band.

- **Obscured AGN.** Thanks to the PHEMTO unprecedented sensitivity reached at high energies ($< 10^{-14}$ erg cm^{-2} s^{-1} in 50-150 keV range), heavily absorbed AGN will be detected not only in the local Universe but also at higher redshift. With PHEMTO sensitivity in the 50-150 keV band we can map the peak of AGN activity up to z∼1–3 for the majority of the population; for the most extreme, rare quasars, we could reach the high-redshift universe (z∼5–6). Current X-ray spectra of AGN at z$\sim 1-3$ are rather poorly known, the uncertainties on their column densities are large and some of them may even be CT. High-energy PHEMTO spectra will easily distinguish between Compton-thin and Compton-thick AGN. For a moderately bright ($L_X = 10^{44}$ erg s^{-1}) Compton thick AGN at redshift z = 1, the absorbing column density can be determined with an error of 10% within a 100 ks observation. This is sufficient not only to confirm the source is Compton Thick, but also to provide a precise measurement of the torus thickness, which is a key metric for studying black hole/galaxy co-evolution at "Cosmic Noon". Moreover, by targeting candidates from previous surveys (XMM-Newton, Chandra, NuSTAR, INTEGRAL) it will be possible to study in detail their broad-band spectra in the $1-400$ keV band for the first time. In a deep 1 Ms exposure, our simulations show that thanks to the PHEMTO sensitivity it will be possible to measure the broad (1-400 keV) X-ray spectral characteristics of the faintest detectable sources up to z $\simeq$ 3 and with a luminosity in the $2-10$ keV energy range less than a few 10^{44} erg s^{-1}, separating unabsorbed, Compton thin and CT objects with high significance, and recovering the intrinsic absorption with less than 20% uncertainty for Compton thin objects.

An additional issue that PHEMTO can address is the investigation of the elusive class of compact obscured nuclei (CONs) which are strong mid-infrared emitters selected in the 14 μm band [Falstad et al. 2021]. The mechanisms that drive the build-up of this nuclear obscuration are still unclear and great efforts are been made in order to understand whether CONs harbour CT AGN. PHEMTO will be able to detect some candidates in the hard X-ray band and help in unveiling their nature. The combination of X-ray and infrared information can be used to measure the number density of highly obscured QSOs. By joining this sample to those obtained from other surveys, we will be able to determine the evolution of the obscured AGN population, a step forward in completing the census of SMBH through Cosmic time.

1.5 The PHEMTO view of particle acceleration, magnetic fields and AGN feedback in galaxy clusters

- **Particle acceleration and electron-ion equilibration in shocks.** Shock fronts generated in cluster mergers are thought to be the physical mechanism that converts gravitational energy into the thermal energy of the ICM. They are also thought to be the cause of the formation of giant, Mpc-size, arc-like structures called radio relics. There are two widely regarded physically plausible heating scenarios for shock heating: either that ions and electrons are both instantaneously strongly heated at the shock front; or that only the ions are heated initially and the electrons then equilibrate through collisions with the ions over a timescale of 100 Myr. Shock fronts can be used to determine the velocity and kinematics of the merger, and to study the conditions and transport processes in the ICM, including thermal conduction and electron-ion equilibration. The temperature profile across a shock front, combined with density-jump based shock Mach number, can disentangle the heating mechanisms of electrons and the microphysical processes at work.

Such measurements are extremely challenging as high shock temperatures ($kT \gtrsim 10$ keV) are poorly constrained by the low effective area above 5 keV of Chandra and XMM, and by the relatively poor angular resolution of NuSTAR. Recent studies cannot resolve the degeneracy between electron shock heating models, or give stringent upper limits on the effective thermal conductivity in post-shock regions [Markevitch et al., 2003, Markevitch, 2006, Russell et al., 2012, 2022]. NuSTAR measurements of the temperature profile across a shock front such as in the recent example of the cluster Abell 665 [Norseth et al., 2025] are providing a key example of what a focusing hard X-ray telescope can achieve. Good angular resolution enables excision of point sources, reducing the CXB fluctuations which are the ultimate limit to determination of the X-ray surface-brightness and electron temperature. **The effective area at**

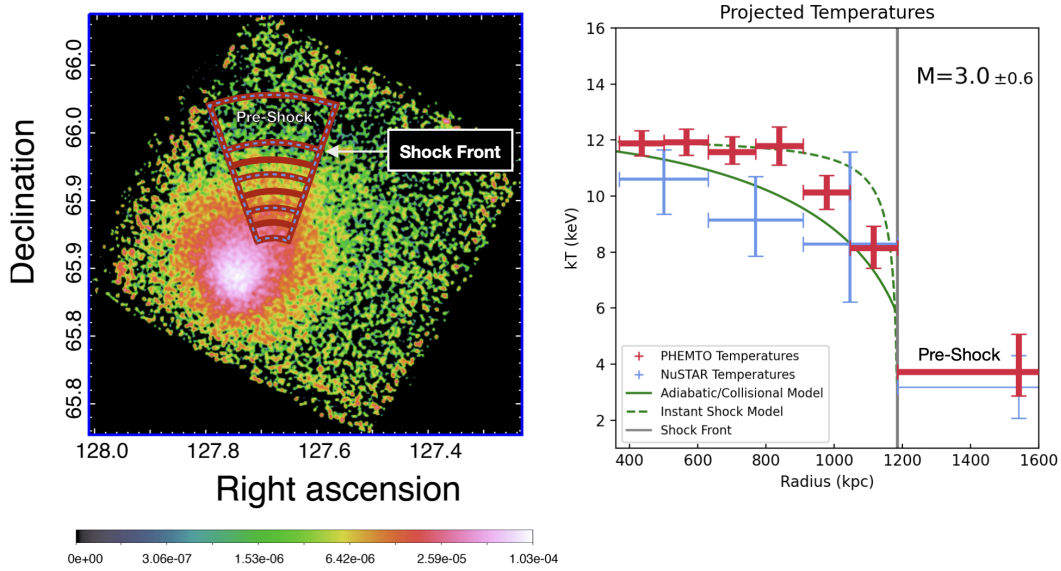


Fig. 1.7: *Left Panel:* An exposure-corrected, background-subtracted, and smoothed 4–25 keV NuSTAR image of galaxy cluster Abell 665 with spectral extraction regions overlaid [Norseth et al., 2025]. The four regions used for the NuSTAR measurements, shown with the dashed blue contours, and the PHEMTO regions, shown with the thick magenta contours, correspond to the data points in the right panel. The shock front is indicated by the arrow. *Right Panel:* Projected NuSTAR temperature profile across the equilibration region and the simulated PHEMTO profile for a 300 ks observation. The predictions for the two different re-equilibration models are also shown, projected along the line of sight accordingly. The shock front is indicated by the solid gray vertical line and a Mach number of 3 is assumed, as measured by the Chandra temperature jump.

high energies and good spatial resolution of PHEMTO will allow these crucial measurements of the ICM physics to be made routinely on a large sample of clusters. Figure 1.7 illustrates the unique discriminating power of PHEMTO with respect to the shock heating mechanism for the case of A665.

By 2040, in conjunction with the increased spatial and spectral resolution of NewAthena, the complex nature of the thermal ICM will be fully unveiled. **A broad band description of this low-beta plasma, taking the ICM spectra to high energies and temperatures, will be critical for further progress in our understanding.**

- **IC emission from radio lobes/cavities.** AGN feedback is a key regulator of the thermodynamics of the intra-cluster gas. In relaxed, "cool core" clusters, relativistic jets from the central AGN release mechanical energy in the form of bubbles and shock fronts in the ICM. These bubbles are visible as X-ray depressions that spatially coincide with radio lobes. X-ray observations with *Chandra* and *XMM-Newton* have revealed tens of such cavities, allowing robust measurements of jet power ($10^{42} - 10^{46}$ erg/s, [e.g., Gitti et al., 2012, Eckert et al., 2021, Ubertosi et al., 2023]). However, the physical content of such bubbles remains poorly constrained. Whether cavities are dominated by relativistic electrons, non-radiating particles (e.g. protons), or magnetic fields is still uncertain [Abdulla et al., 2019], and this uncertainty propagates directly into estimates of jet composition and coupling to the ICM.

IC emission (mostly from scattering of CMB photons) provides a powerful probe of cavity composition. While the lobe synchrotron emission depends on both the electron population and the magnetic field, IC scattering of the CMB depends only on the relativistic electrons. The ratio of radio to X-ray flux densities, $R = S_\nu^r / S_\nu^X$, therefore constrains the magnetic field strength through $B^{\alpha+1} \propto R$, where α is the spectral index [e.g., Hardcastle and Croston, 2010, Mernier et al., 2019]. Once B is known, the magnetic pressure p_B and the electron pressure p_{e^-} can be estimated, allowing the total non-thermal pressure within the lobes, $p_{\rm nth}^{\rm in} = p_{e^-} + p_{p^+} + p_B$, to be compared with the external thermal pressure of the surrounding ICM, $p_{th}^{\rm ext}$.

Assuming pressure balance between the radio lobes and the ICM ($p_{\rm nth}^{\rm in} \simeq p_{\rm th}^{\rm ext}$), IC measurements thus provide direct constraints on the contribution of non-emitting protons p_{p^+} to the pressure equilibrium, and thus on the proton content of AGN jets [e.g., Hardcastle and Croston, 2010, Gitti et al., 2012]. So far, such studies of IC from X-ray cavities have been hampered by instrumental limits: in the 0.5–7 keV band of *Chandra* and *XMM-Newton*, any IC emission from radio lobes is overwhelmed by thermal ICM emission projected onto the bubbles. This is why detections remain limited to a handful of exceptional systems (e.g., Cygnus A, [Snios et al., 2018, de Vries et al., 2018]; Hercules A, [Ubertosi et al., 2025]).

PHEMTO would open a new window on this science: at $\geq$ 10 keV, the emissivity of thermal bremsstrahlung rapidly declines, while IC emission from radio lobes is expected to dominate. The sensitivity and $1-400$ keV coverage of PHEMTO would enable detection of IC in samples of clusters – specifically those hosting large, well-characterized X-ray cavities with diameters $\geq 20''$, where the bubble volumes and relativistic electron populations are already constrained by deep radio and X-ray observations (e.g. archetypal cavity systems such as Perseus, Hydra A, or Abell 2052, among others; see [McNamara and Nulsen, 2007, Gitti et al., 2012]). Simulations using the PHEMTO low- and high-energy detector response files indicate that a 50 ks exposure would enable detection of IC emission down to levels of $\sim$4 nJy in the X-ray cavity in Hydra A (one order of magnitude lower than the current upper limit), while keeping the relative uncertainties on the photon index and normalization of the power-law component below $\sim$30%. Consequently, such measurements of IC flux densities within cavities would allow the contribution of non-radiating particles to cavity support to be quantified with relatively modest exposures of selected galaxy clusters.

PHEMTO's $10''$ angular resolution would even allow to *spatially resolve* the IC X-ray emission across extended radio lobes, enabling the first maps of magnetic field strength within X-ray cavities. Cool-core clusters such as Hydra A, Cygnus A, Hercules A, or MS 0735.6+7421 are suited to this study due to their radio bright, large-angular-size radio lobes and well-characterized X-ray cavities [e.g., McNamara and Nulsen, 2007, Gitti et al., 2012]. Recent work has reported a detection of IC emission from the radio lobes in Hercules A using *Chandra* [Ubertosi et al., 2025]. For the fainter western lobe (radius of $31''$), the IC flux density at 1 keV is $\sim$ 8.9 nJy within a semicircular extraction region. Given the $\sim 10''$ PSF of PHEMTO, the half-lobe could be spatially resolved into $\sim$ 4 independent regions; within each, the expected IC flux density would be of $\sim$ 2.2 nJy. Simulations using the PHEMTO low- and high-energy response files indicate that a $\sim$ 200 ks exposure would allow the IC detection in each region with $\sim$ 25% relative uncertainty on the power-law normalization, while simultaneously accounting for the contribution of projected thermal emission in the 1–10 keV band. Crucially, the extension of PHEMTO to $\sim$ 50–200 keV, where the IC component dominates, is key to separation of these components. The resulting uncertainties on the IC normalization imply sensitivity to spatial variations in magnetic field strength at the level of a factor of $\sim$ 2–3 across the lobes. With deep PHEMTO exposures, we could thus obtain the first maps of IC-based magnetic field strengths within X-ray cavities.

- **IC emission from diffuse radio sources in clusters and magnetic field measurement.** The presence of relativistic particles and magnetic fields in the intra-cluster medium (ICM) of galaxy clusters has been established by a number of observations at radio frequencies [Feretti et al., 2012, van Weeren et al., 2019]. Excluding individual galaxies, cluster diffuse radio sources can be classified into halos (large-scale diffuse, extended, low surface brightness, steep-spectrum synchrotron sources) and relics (localised sources in the cluster periphery).

Halos are produced by synchrotron emission from relativistic electrons spiralling around $\sim \mu$G magnetic fields. They are ubiquitous in clusters that show indications for merging activity as indicated in X-rays or optical [Buote, 2001, Cassano et al., 2010, Govoni et al., 2004, Girardi et al., 2011]. Haloes are thought to be generated by the dissipation of the kinetic energy from cluster mergers through turbulence [Brunetti and Jones, 2014]. The similarity of the halo radio morphology to that of the thermal ICM [Govoni et al., 2001a] and the correlation between the total radio halo power $P_{1.4GHz}$ at 1.4 GHz and the cluster total X-ray luminosity L_X [Giovannini et al., 2009, Kale et al., 2013], suggest a connection

between the thermal and relativistic plasmas.

Similarly to the case of radio lobes discussed above, relativistic electrons that produce radio halos and relics also produce hard X-ray emission by IC scattering photons from the CMB. However, this IC signal it is difficult to detect because it is swamped by thermal photons below 10 keV. Nonetheless, the IC emission should dominate at higher energies, where the bremsstrahlung continuum falls off exponentially. For a power-law energy distribution, the IC/synchrotron flux ratio gives a direct measurement of the average magnetic field strength in the ICM. The results of IC searches with X-ray satellites remain controversial. *NuSTAR* [Harrison et al., 2013], the first satellite with imaging capabilities in the hard X-ray band (3-79 keV) made pioneering observations of merging clusters [Wik et al., 2014, Gastaldello et al., 2015, Cova et al., 2019, Rojas Bolivar et al., 2021, 2023, Tümer et al., 2023] clearly revealing that the average magnetic field in radio halos clusters is typically closer to $\sim 1\mu$G than the 0.2 μG implied by past detections. However, a definitive detection of IC emission in hard X-rays remains elusive.

While the majority of the diffuse sources in the ICM have been historically detected in clusters at redshift $z < 0.4$, highly sensitive observations with new generation radio interferometers, mainly LOFAR and MeerKAT, are starting unveiling a population of high-z clusters, even beyond $z > 1$ [Di Gennaro et al., 2021, Sikhosana et al., 2025, Phuravhathu et al., 2025, Di Gennaro et al., 2025]. These clusters are particularly suited for the search for IC from the ICM, due to the stronger IC losses at high-z. Detecting IC (or providing limits to) in these systems can provide important information on the magnetization of the ICM, and thus on how magnetic fields are amplified in clusters in the early Universe. PHEMTO will be able not only to detect, but to **map** with spatially resolved spectroscopy the IC emission, allowing reconstruction of the spatial distribution of the relativistic electrons and the magnetic field, in a way complementary to the Rotation Measure constraints provided by SKA. RM is sensitive to line-of-sight component of the magnetic field weighted by the thermal electron density while IC is sensitive to the average value of the field in the emitting volume. **PHEMTO will enable reconstruction of the physical properties of the non-thermal particles of galaxy clusters to the same level of accuracy likely achieved by NewAthena for the thermal ICM.**

Science Goals	Science Objectives	Observing Approach	Required Measurements
Understand the role of magnetic field in cosmic accelerators	• SO1: The role of magnetic field in SNR and PWN • SO2: Magnetars as powerful cosmic accelerator	• Guest Observers program • TOO proposals • Synergy with SKA, NewAthena, CTAO	• sensitivity • field of view • energy range • angular resolution • polarization
Probe the nature of accretion/ejection in accreting black hole and neutron star systems	• SO3 : Accretion in Black Holes binaries of all mass scales • SO4 : Accretion in magnetized Neutron Stars	• Guest Observers program • TOO proposals • Synergy with SKA, NewAthena, CTAO	• energy range • timing • polarization
Constrain explosion physics in supernovae and their remnants	• SO5 : Discriminate between progenitor channels for SN Ia • SO6 : ^{44}Ti studies for supernova remnants	• TOO proposals • Guest Observers program • Synergy with NewAthena	• spectroscopy • sensitivity • angular resolution • field of view
Describe the structure and demography of obscured AGN	• SO7 : Study the Cosmic X-ray background • SO8 : Extend the known AGN population and probe their evolution up to a redshift z=3 • SO9 : Analysis of Compton Thick AGN	• Guest Observers program • TOO proposals • Synergy with SKA, NewAthena	• energy range • sensitivity • field of view
Determine particle acceleration at shocks and the non-thermal energy content of clusters of galaxies	• SO10 : Study particle acceleration and electron-ion equilibration in shocks • SO11 : Detect and map IC emission from diffuse radio sources and AGN cavities in clusters	• Guest Observers program • Synergy with SKA, NewAthena	• sensitivity • field of view • angular resolution • low background

Table 1.1: PHEMTO Science traceability matrix

2. Scientific and Mission Requirements

To address these main scientific objectives, we have built the PHEMTO traceability matrix (Table 1.1) and derived the PHEMTO science requirements. With such characteristics, PHEMTO will address many additional questions in other astrophysical domains and, without any doubt, many serendipitous objectives and questions which may be opened in the next decades.

We show in Figure 2.1 a comparison of PHEMTO's expected performance with other high energy missions. PHEMTO's large energy range and polarization capacities encompasses several previous missions like NuSTAR, INTEGRAL/IBIS and ASTROSAT. Thanks to the Laue lens, PHEMTO will be the first X-ray focusing telescope above 80 keV which explains the large gap in angular resolution and sensitivity compared to survey telescopes such as INTEGRAL/IBIS coded mask instrument and COSI Compton telescope. Performance for INTEGRAL NuStar, ASTROSAT and COSI come from [Lebrun et al., 2003, Harrison et al., 2013, Vadawale et al., 2015, Tomsick et al., 2024].

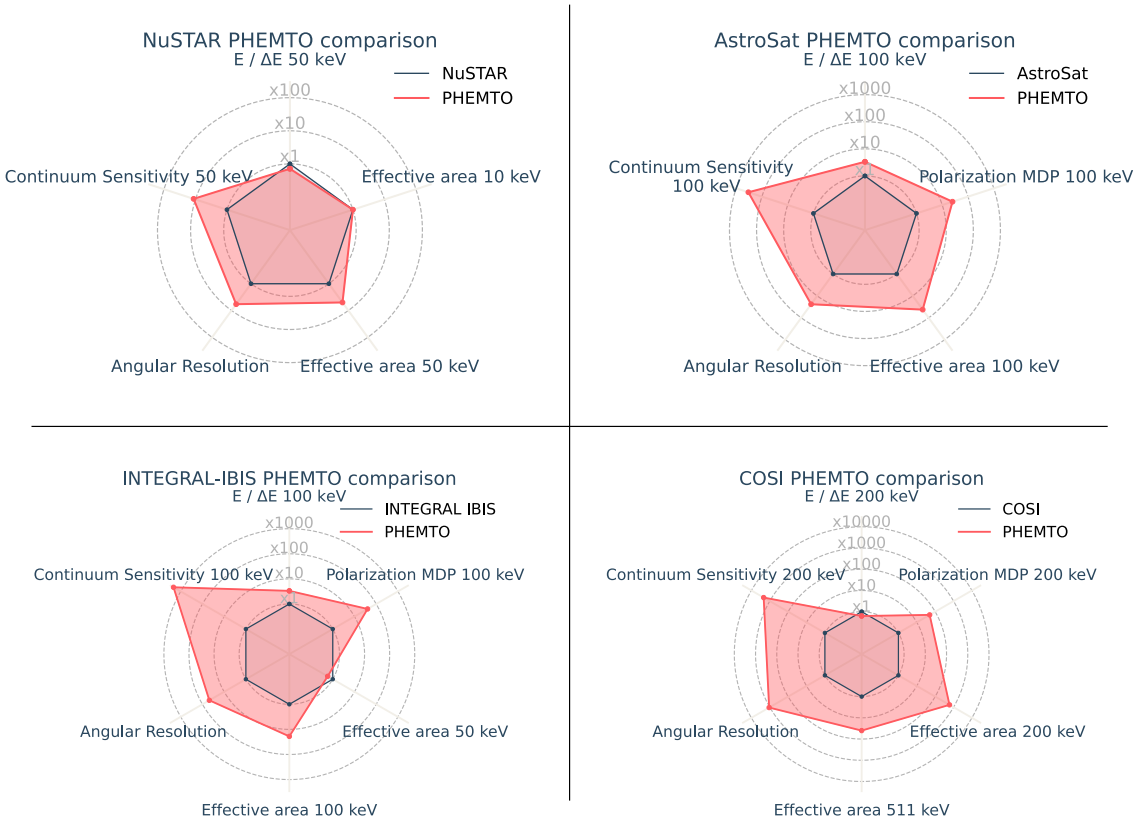


Fig. 2.1: PHEMTO performance comparison with the high energy missions NuSTAR, INTEGRAL-IBIS, COSI and ASTROSAT/CZTI. All pointed telescopes assume an exposure of 1 Ms; for COSI a 2-year survey is assumed.

Table 2.1: PHEMTO scientific requirements

	Parameter	**Value**	**Science drivers**
SR-1	Energy band	1 - 400 keV	SO1, SO2, SO3, SO4 SO7, SO8, SO9
SR-2	On-axis continuum sensitivity (1 Ms, $\Delta E = E/2$)	2×10^{-15} erg cm^{-2} s^{-1} @ 10 keV 3×10^{-15} erg cm^{-2} s^{-1} @ 100 keV 2×10^{-13} erg cm^{-2} s^{-1} @ 400 keV	SO1, SO2, SO7, SO8, SO9, SO10, SO11
SR-3	On-axis line sensitivity (100 ks, $\Delta E \sim E/50$)	10^{-7} ph cm^{-2} s^{-1} @ 100 – 300 keV	SO5, SO6
SR-4	Minimum detectable polarisation (100 ks, 100 mCrab)	10 % @ 30 keV 5 % @ 200 keV	SO1, SO2, SO3, SO4
SR-5	Field of view	10 arcmin @ 10 keV (mirror) 5 arcmin @ 100 keV (lens)	SO1, SO2, SO5, SO6, SO7, SO8, SO9, SO10, SO11
SR-6	Angular resolution (H.E.W.)	$\approx 10''$ @ 10 keV (mirror) $\approx 25''$ @ 200 keV (lens)	SO1, SO2, SO5,SO6, SO10, SO11
SR-7	On-axis effective area	500 cm^2 @ 10 keV (mirror) 16500 cm^2 @ 100 keV (lens)	SO1, SO2, SO3, SO4, SO5, SO6, SO7, SO8 SO9, S1O, S11
SR-8	Spectral resolution	E/ΔE = 50	SO3, SO4, SO5, SO6 SO8, SO9
SR-9	Detectors background	$\leq 10^{-4}\,\mathrm{cts\,s^{-1}\,cm^{-2}\,keV^{-1}}$	SO5, SO6, SO10, SO11
SR-10	Absolute timing accuracy	50 μs	SO3, SO4
SR-11	Localisation accuracy	$\approx 5''$	SO1, SO2, SO8, SO9 SO10, SO11
SR-12	Mission duration	5 years (nominal) + 3 years (extended)	SO5, SO6, SO7, SO8 SO10, SO11

2.1 Scientific requirements

The PHEMTO scientific requirement are given below and are detailed in Table 2.1.

SR-1 broad-band coverage starting at 1 keV, is necessary to constrain and cleanly separate different continuum and absorption spectral components. Extending the band up to 400 keV is important to cover the ^{44}Ti line and to constrain efficiently the high-energy cut-offs in AGN's and black holes spectra.

SR-2 continuum sensitivity needed to make polarimetry mapping of several sources (neutron stars, magnetars, black holes, bright AGN, ...). It is also needed in order to accumulate meaningful spectra in variable objects in a duration comparable to their dynamical time-scales.

SR-3 line sensitivity needed to detect the radioactive lines emitted by supernovae after the explosion and to study their remnant.

SR-4 minimum detectable polarisation ensures accurate measurements and mapping of extended sources such as SNRs and PWNs and open sensitive window for precise measurement of polarization in binaries.

SR-5 field of view a large FOV is needed for deep surveys targeted to the resolution of the Cosmic X-ray Background, as well as for the study of acceleration in extended sources (clusters of galaxies, supernovae remnants, jets), and of binary systems in close by galaxies.

SR-6 angular resolution necessary for both reaching the required sensitivity and avoid source confusion. Also necessary for resolving features like radio-galaxy hot spots and lobes and supernovae remnants, and to avoid confusion in complex and crowded regions like the Galactic Centre, star-formation regions etc...

SR-7 On-axis effective area minimum telescope effective area in order to meet the above sensitivity requirement and to ensure an adequate spectral capability across the entire energy band.

SR-8 spectral resolution needed to separate different absorption and emission spectral components at low energy, including the Fe line, and characterize at high energy the ^{56}Ni, ^{57}Ni, and ^{44}Ti lines in supernovae and supernovae remnant.

SR-9 detectors background ensures the required line and continuum sensitivity on axis for point sources, and allows sensitive mapping of extended sources with low surface brightness.

SR-10 absolute timing accuracy needed to perform absolute phase determination of e.g. ms pulsars, and compare with radio data, requires an absolute timing accuracy of at least 100 µs (50 µs goal), obtained after on ground processing.

SR-11 localisation accuracy needed to search for correlation with other wavelengths and to characterize hot spots in extended sources.

SR-12 mission duration the minimum to ensure that all the scientific goals of the mission are met. The mission being an observatory, and the potential targets being very numerous and for some of them very variable, an increase of the mission duration will imply a science return increase.

3. Payload Description

3.1 Measurement Concepts

The envisioned technical implementation for the PHEMTO observatory is to have two independent focusing systems looking simultaneously at the same sky. One more efficient at low energy, based on focusing mirrors working in reflection conditions, and the other one at higher energies, based on the Laue lens technique (see [Takahashi, 2005, Frontera et al., 2013]). These optics, located at the end of an extendable boom, will focus hard X-rays/gamma-rays on the same point of a Compton telescope, located 20 meters away on the satellite platform. The focal length will be then 20 meters. The Compton telescope will be composed of two stacked detectors, one in Silicon, the other in CdTe. According to their energies, photons will be detected either by the Silicon detector, by the CdTe ones or both. In case of

double Compton detection, the Compton kinetics will enable us to measure the incident photons linear polarisation state. These measurements will be detailed in the following parts below.

3.2 System overview

The PHEMTO payload consists of a single telescope with 20 meter focal length. An overview is presented in Figure 3.1. The system can be decomposed in four main assemblies:

- The optical system, based on a dual focusing optics.
- The detector system, based on a dual focal plane.
- The telescope system, that insures the alignment of the two previous systems.
- The instrument control unit, that manages the operation of the instrument and the science data processing.

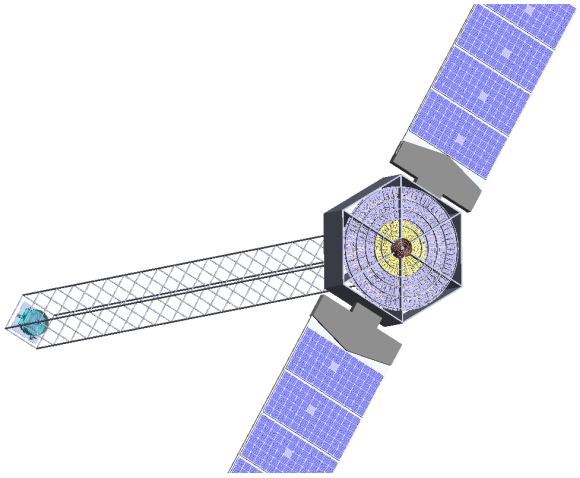


Fig. 3.1: CAD view of the PHEMTO full telescope. The central red part is the mirror, the cyan part is the Lens part made with Silicon tiles and the yellow part is the ones made with Germanium tiles. We see the detector payload in blue at the end of the boom. The platform behind is not shown.

The following sections will describe, for each assembly, the design drivers, the physics principles, a baseline design, the design justification by performance analysis and the relevant heritage to build such systems.

3.3 Optical system

3.3.1 Assembly description

The mirror optic will be made of Ni-electro-formed full shell mirror substrates, leveraging the heritage of XMM-Newton and coated with Pt/C and W/Si multi-layers for an effective energy range of $1-80$ keV. Laue lenses exploit the crystal diffraction in transmission configuration (Laue geometry) and is made of a large number of crystal tiles which are disposed in such a way that they concentrate the incident radiation onto a common focal spot. The lens crystals will be of two types, Silicon in its external parts to focus lower energy gamma-ray and Germanium in its internal parts to focus photons up to 400 keV. The mirror will be accommodated within the Laue lens inner radius, which will have an outer radius of 150 cm, joining the two focusing optics into a single unit.

3.3.2 Mirror system

The mirror optic module will be made at Media-Lario (Italy) of Ni-electro-formed full shell mirror substrates. This technology has been successfully used for the Ni gold-coated X-ray mirrors of BeppoSAX, Swift, XMM-Newton and eROSITA. It makes use of Wolter-I paraboloid-hyperboloid mirror shells, nested together into their case through a spider-like structure. The current design foresees 57 mirror shells with decreasing diameters from 500 to 296 mm, 20 m focal length, and a 330 + 330 mm parabola-hyperbola total length. The mirror thickness will vary in proportion to the radius from 0.547 mm to 0.324 mm to provide optimal structural stiffness, while keeping the total module mass within 250 kg, including structures. The reflective coating will be based on multilayer structures optimized for high-energy performances, with shells coated with W/Si at the outermost radii and Pt/C for the innermost shells. This solution will ensure an effective area on-axis close to 500 cm^2 up to 10 keV and slowly decreasing up to 70-80 keV, where the effective area is close to 70 cm^2. Based on the experience gained with eROSITA

and the superior surface smoothness with respect to previous missions such as XMM, we expect an excellent angular resolution (HPD) of $\sim 10''$ at 10 keV, and, due to the extremely slow increase of the X-ray scattering relevance due to the very low roughness, an HPD lower than $25''$ for energies below 80 keV.

3.3.3 Laue Lens system

Laue lenses represent one of the most promising solutions for achieving unprecedented sensitivity in the hard X-ray and soft gamma-ray domain ($\sim$ 100 keV–MeV), enabling breakthrough science in high-energy astrophysics. Their development has been pioneered by a few groups in Europe, including University of Ferrara (Italy) with contribution from INAF/OAS Bologna, the Institut de Recherche en Astrophysique et Planetologie, Toulouse (France) with the contribution of industrial partners.

The Laue lens concept is based on the diffraction of photons in transmission geometry by suitably oriented crystal planes. By arranging a large number of crystals according to precise geometrical laws, photons can be focused onto a detector, significantly enhancing the signal-to-background ratio compared to non-focusing instruments. Despite its strong scientific potential, the realization of a space-qualified Laue lens poses significant technological challenges. These challenges can be grouped into two fundamental and interdependent tasks:

- The production of high-quality diffractive crystal elements;
- The precise mounting of thousands of such elements into a common supporting structure.

These two aspects define the technological roadmap for the PHEMTO mission.

Crystal development

The first requirement is the availability of several thousands of crystal elements with well-controlled optical properties. Each crystal must provide high diffraction efficiency in the selected energy range and exhibit a controlled angular spread of the diffracting planes. A key parameter is the curvature radius in the case of bent crystals, which directly determines the focusing capability. Bent crystals have demonstrated clear advantages over flat mosaic crystals, as they allow improved reflectivity and better photon concentration. However, achieving a precise and reproducible curvature remains a critical aspect. In addition to curvature, the crystals must ensure:

- uniformity across large batches;
- stability of their properties over time;
- minimal lattice defects and controlled mosaicity;

Significant progress has been achieved in recent years, particularly with Silicon and Germanium crystals. Nevertheless, scaling up to mass production while maintaining tight tolerances is still a challenge and requires continued development.

Assembly / mounting strategy of the Laue Lens

The second major task concerns the integration of thousands of crystals into a common support structure. Each element must be positioned and oriented with arcsecond accuracy in order to satisfy the Bragg condition and ensure proper focusing. Typical requirements involve angular accuracies of a few arcseconds and positional tolerances at the level of tens of micrometers. Even small deviations can degrade the point spread function and reduce the overall sensitivity of the instrument.

The baseline solution adopted for PHEMTO is based on the bonding of diffractive crystals onto a common frame made of low thermal expansion materials (quartz or Zerodur substrate). The crystals are attached using low-shrinkage space-qualified adhesives, ensuring mechanical robustness and compatibility with the space environment. This approach allows precise alignment during integration while maintaining structural stability. This technique has been validated through the realization of small-scale prototypes

([Ferro et al., 2024]). On this basis, the technology can be considered at a TRL of about 3/4, corresponding to laboratory validation of the concept. In Figure 3.2 is shown a prototype of petal made of Silicon crystals in the preparation phase, developed at the LARIX facility of the University of Ferrara. With this technology we have reached up to now an accuracy at level of a module made of tens of crystals of the order of a few arcmin.

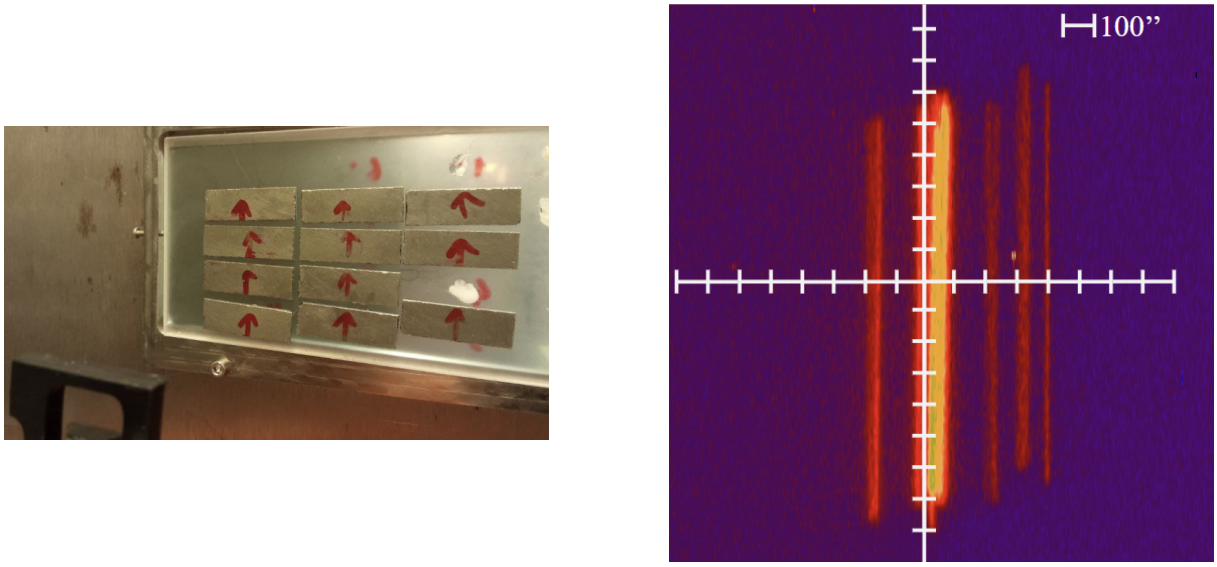


Fig. 3.2: Left: prototype of Laue lens made of 11 Silicon bent crystals glued on a quartz glass substrate. Right: Alignment capability of the 11 Silicon crystals with respect to the target.

Ongoing R&D activities

To increase the maturity of the technology, several R&D activities are ongoing at both national (ASI) and European (ESA) level, addressing both crystal production and mounting strategies. At the University of Ferrara, within the OTELLO project (ESA GSTP), efforts are focused on:

- investigating anodic bonding techniques to reduce positioning uncertainties and improve stability;
- developing high-precision Silicon and Germanium crystal samples with controlled properties;

As the substrate will be accurately prepared with a proper finishing of the surface such that its curvature matches the focal length F of the Laue lens (F = R / 2), the anodic bonding technique will be used both to fix the crystals to the substrate and to induce an accurate curvature to the sample.

For the fabrication of high-precision silicon-based optical elements, cosine (The Netherlands) has developed a novel approach to producing Laue lens elements, called Silicon Laue Components (SiLC). This method relies on scalable mass-production techniques using commercially available silicon wafers and builds on more than two decades of work by the European Space Agency, cosine, and their partners on Silicon Pore Optics (SPO) [Barrière et al., 2022], the mirror technology developed for the Athena X-ray Observatory.

SiLC are fabricated from monocrystalline silicon plates cut from double-sided super-polished wafers [Barrière et al., 2023]. The plates are assembled by direct silicon bonding, eliminating the need for adhesives. The plates are stacked against a precision mandrel that defines the optical figure. The extremely small thickness variation of the plates, combined with the absence of intermediate bonding layers, ensures accurate replication of the mandrel geometry. The result is a self-supporting stack of elastically deformed crystalline plates reproducing the mandrel figure. Each plate incorporates a small wedge angle so that successive plates diffract towards a common focal point. Sagittal focusing is provided by the sagittal curvature of the plates, while radial focusing is achieved through the inter-plate wedge. The diffraction

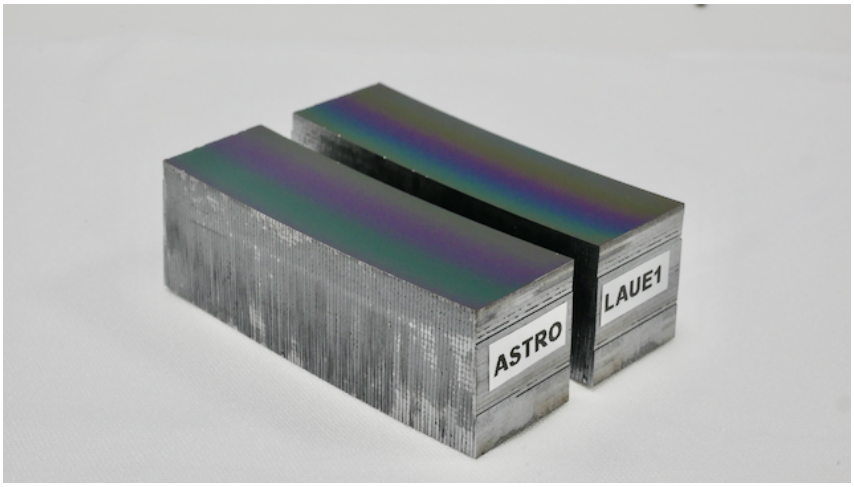


Fig. 3.3: Two demonstrator SiLC fabricated within the ESA-funded project ASTROLAUE1 one of which was characterized at LARIX in a 120 keV beam and exhibited excellent performance.

bandpass is defined by the meridional curvature imposed by the mandrel. The resulting SiLC forms a self-standing composite crystal of large dimensions, up to 120 mm in the azimuthal direction and 40 mm in the radial direction, with both sagittal and meridional curvatures integrated into the structure. The combination of these curvatures with the inter-plate wedge provides strong focusing capability while remaining largely independent of both the element size and the diffraction bandpass. The manufacturing process and performance have recently been demonstrated through successful testing of a SiLC demonstrator 3.3, measured at 120 keV at the LARIX facility of the University of Ferrara, Italy. SiLC benefit directly from the technological heritage of SPO, which has reached TRL of 6 through its development for the NewAthena X-ray Observatory programme led by the European Space Agency.

Laue lens design

Silicon crystals cover the lower part of the energy passband of the Laue lens system, from 50 keV to 100 keV. The high energy part, up to 400 keV, will be covered by Germanium crystals. The lens is designed for a focal length of 20 m. The lens radius spans from 646 mm to 1800 mm, with the outermost ring tuned to 50 keV and the innermost ring to 400 keV.

The Si/Ge arrangement of the lens is optimised in the framework of the GSTP Otello project.Indeed, the lens proposed in that framework is made of bent crystal tiles of Si(111) and Ge(111), with 40 m curvature radius (within 5% uncertainty). The crystal tiles have been assumed to have a 30×10 mm^2 cross section, and an optimized thickness vs. energy for maximizing the diffraction efficiency. The expected effective area and PSF of the entire PHEMTO lens has been simulated and validated by measurements on several subsets of crystals. Thanks to the use of bent Si(111) and Ge(111) crystals tiles, an energy passband 50–400 keV should be achievable.

If the solution based on SiLC will be chosen for the lens low energy response, each optical element is composed of 40 plates, each 0.775 mm thick, resulting in a radial extent of 31 mm per element. With a 2 mm gap between adjacent rings and a 10 mm gap between neighbouring SiLCs within a ring the low energy part of the Laue lens results to have a filling factor of 85%. The prototype optical element effective area peaks at approximately 580 cm^2 at 80 keV These values are obtained assuming optical elements with a diffraction bandpass of 60 arcsec. The resulting point spread function (PSF) is strongly peaked, enabling source localisation at the level of 10 arcsec. This characteristic suggests that larger SiLC bandpasses could be considered. Increasing the bandpass would increase the effective area while broadening the PSF wings, but without significantly degrading the source localisation capability.

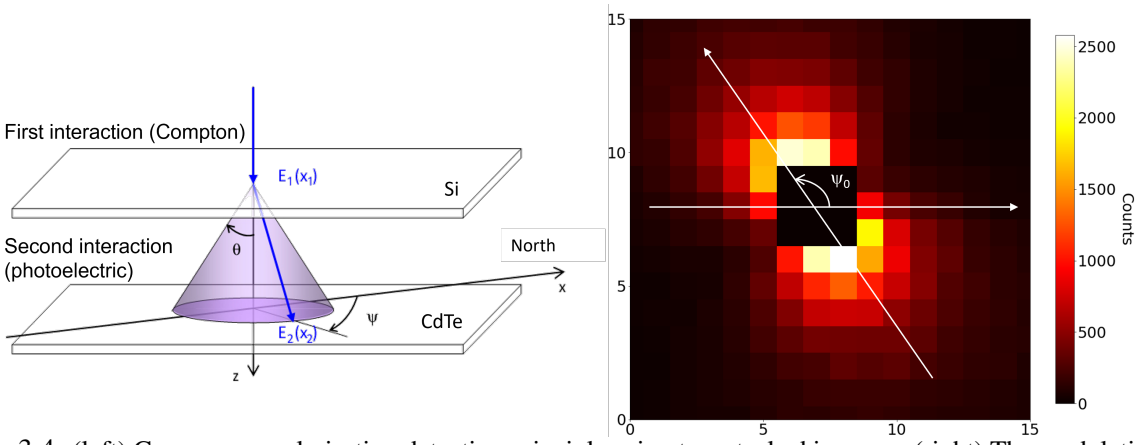


Fig. 3.4: (left) Gamma-ray polarisation detection principle using two stacked imagers. (right) The modulation curve is built in ψ_0 bins, from the count map on the second detector (that ignores the central pixels to limit ambiguity due to charge sharing).

3.4 Detector system

3.4.1 Assembly description and subsystem definition

The detector system shall provide photon detection with imaging, spectral and polarimetric capabilities (as defined in Table 2.1) with a good background rejection to reach the telescope sensitivity requirements. It will consist in a Focal plane assembly (FPA) placed in front of a Filter wheel assembly (FWA) and a Collimator assembly. To operate the FPA, a Detector thermal control system will be implemented as well as a Detector Control Electronics (DCE) system. All these elements will be placed on the platform.

3.4.2 Focal plane assembly

The focal plane assembly consists of the stacking of a Low energy Detector (LED) and a High energy detector (HED) surrounded by an anticoincidence system. This dual focal plane allows to exploit the high spectral resolution capability of silicon at low energy (150 eV FWHM at 6 keV) with the high spectral resolution capability of Cd(Zn)Te in the gamma-ray range (1.5 keV FWHM at 100 keV). Above 25 keV, this configuration favours the Compton scattering of the gamma-rays in the silicon layer (low Z material) and the full absorption of the scattered photon by photoelectric effect in the CdTe layer (high Z material). In case of a light linear polarisation, the azimuthal count distribution of the secondary photons in the CdTe plane is not symmetrical but shows a modulation like in Fig. 3.4:

$$N(\phi) = S[1 + \mu \; cos \; 2(\phi - \phi_0)] \tag{3.1}$$

The modulation amplitude will give the fraction F of polarisation $F = \mu_0/\mu_{100}$. The modulation phase ϕ_0 will give the angle of polarisation α: $\phi_0 = \alpha + \pi/2$. μ_{100} is a characteristic parameter of the polarimeter, corresponding to the modulation due to a 100 % polarised incoming beam.

LED description

The LED is a monolithic silicon pixel detector. It will consist of an array of 256×256 DEPFET (DEpleted P-channel MOSFET) macropixels with a pixel size of $250 \times 250 \; \mu m^2$ each, integrated on a high resistivity silicon bulk with 450 μm thickness. The combined detector / amplifier structure DEPFET uses a p-channel readout MOSFET for each pixel to sense the presence of signal charge. The bulk generated signal charge is collected within an artificially created potential minimum underneath the gate of this MOSFET, and its presence modulated the channel conductivity. The comparably compact DEPFET readout cell is surrounded by concentric drift rings to implement the large pixel area while maintaining the low readout capacitance. For the PHEMTO LED, the pixels will have three drift rings, with the outermost drift ring

being common for all pixels. A comparable design, but with a different readout FET structure, has been implemented for the X-ray detectors for the MIXS-Instrument on BepiColombo ESA mission [Treis et al., 2010]. The design of the readout DEPFET cell itself will be a linear gate device similar to the designs created for the ATHENA pixel detector. The final sensitive area of the detector will be 64×64 mm^2.

The LED is read out in 1-fold column parallel rolling shutter mode, using 4 specimen of the analog amplifier / filter ASIC VERITAS II. A row readout time of 2.5 μs is targeted, which yields a total exposure time for each frame of 640 μs. The IC produces a serial analog output signal stream, which requires digitization in the backend. Readout is controlled by in total four Switcher ICs of type SWITCHER A. Both readout and control ICs were designed primarily for the ATHENA mission based on earlier versions developed for MIXS, and are qualified for spaceflight use. The detector can be read out flexibly and allows setting ROI windows on the fly with accordingly faster readout repetition rate for the observation of bright targets or in case high time resolution is required.

As for polarimetric applications and background suppression, the timing capabilities of the DEPFET matrix are not sufficient. Therefore, fast timing capabilities with limited localization are implemented by evaluating the hole signal arriving on the entrance window side (aka backside) of the LED detector with high time resolution. For this purpose, the backside contact is segmented, and the charge signal is read out by a suitable multi-channel CSA circuit. The design foresees a segmentation into a total number of 64 strips, each covering in total 8 pixel columns and half of the pixel rows. For readout of the complete backside, two IDeF-X chips will be required.

Readout electronics (i.e. ASICs and required passives) for both spectroscopy and fast timing will be located on a common readout circuit board, which is mounted together with the detector in the focal plane. Connection between ASICs and detector is established via Aluminum wedge-wedge wire bonding. Connection between the readout circuit board and the instrument periphery will be established by a number of custom-designed flexlead circuits, as illustrated in Fig. 3.5.

HED description

The HED layer will be based on Cd(Zn)Te pixel detectors. The fine pixel pitch of 250 μm will allow to well sample the point spread function and get a fine localisation of the secondary photons to enhance the polarimetric capability. The detection layer is thought as an mosaic of large CdTe crystals of 24×24 mm^2. The baseline design assumes a 2-mm thick CdTe detector with a Schottky contact on the pixel side from Siemens Acrorad Ltd. as the best flight heritage (Solar Orbiter/STIX, SVOM/ECLAIRs) of high quality material for charge transport and low leakage current, both critical parameters for spectroscopy. This is a conservative approach for performance. In the early phase of the project, thicker CdTe or CdZnTe detectors from various procurement sources will be considered to increase the detection efficiency above 200 keV. The CdTe crystals are part of hybrid detectors based on the 3D technology of the 3D PLUS company. The 96×96 pixel CdTe detector is point to point connected to a front-end layer made of the stitching of 2×2 ASICs with 48×48 spectroscopic channels. A 3D packaging integrates this front-end layer and other layers for analog to digital conversion (ADC) and filtering components, so that the hybrid detector is actually a full detection unit with a digital interface. With a mosaic of 3×3 detection units, the HED fulfills both requirements of angular resolution with a good sampling of the PSF (1 mm HEW in the detection plane) and field of view. This detection plane can be thought as three independent ladders with individual supplies to ensure sectoring operation and reliability.

ACD description

The anti-coincidence system is composed of a passive shielding to stop high energy particles or photons below 100 keV and an active shielding to reject by time coincidence the high energy particles or photons passing out of the field of view and depositing charges in the LED or HED. The passive shield is a graded Z-shield as a sandwich of material (Pb, In, Sn, Al, C) to prevent from emitting X-ray fluorescence towards

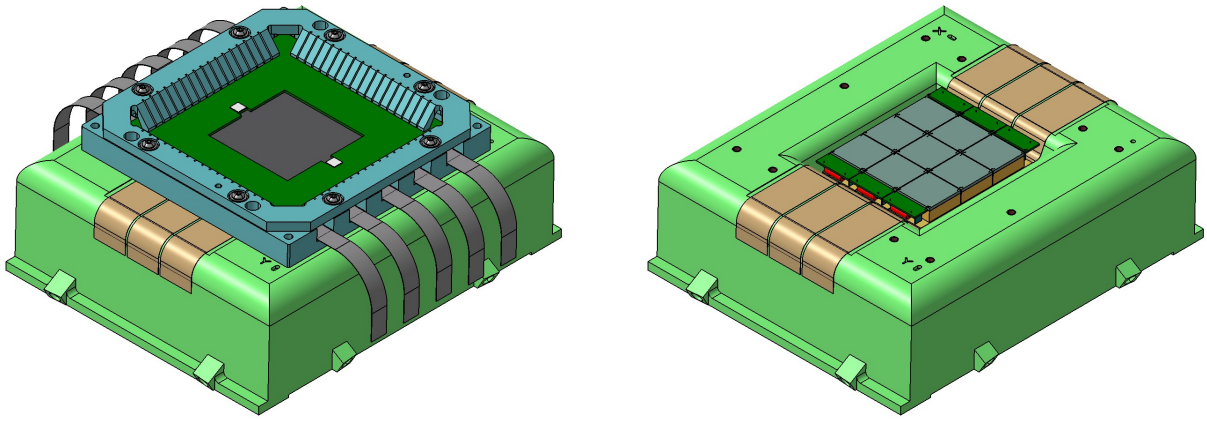


Fig. 3.5: Assembly of two detection layers (Left) The low energy detector assembly, with the visible trigger side on the top (Right) The high energy detector just underneath, made of 3 x 3 detection units to have a central detector for punctual sources

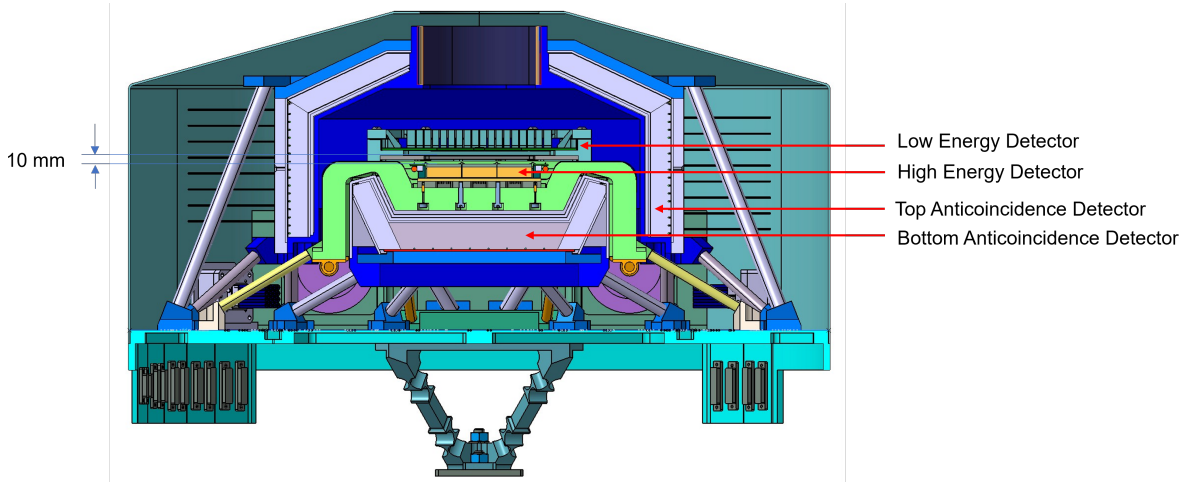


Fig. 3.6: Cut of the PHEMTO Focal plane assembly showing the three detectors and the enclosure. This preliminary design is a refurbishment of the Simbol-X/Cospix mission with an upgrade of the LED and HED. The distance between LED and HED can be optimised to be as low as 10 mm.

the focal planes. The active shield is a assembly of plastic scintillator plates coupled to photodetectors. The baseline for the scintillator is an Eljen EJ-208 polyvinyltoluene scintillator, which will be completely wrapped, except for the two light output windows, in a 3M Vikuiti reflecting layer. The baseline for photodetection is the use of silicon photomultipliers (SiPM) as low mass high gain solutions and emerging flight heritage (EirSat-1, GRBAlpha [Pál et al., 2023]) The front-end ASIC for the SiPM readout, based upon the IDEAS/SIPHRA ASIC will be part of the focal plane assembly. As it could be seen in Figure 3.6, the ACD will be composed of thirteen plates, five for the bottom anticoincidence and eight for the top ones. The optical signal from these plates will be read by two SiPM each for redundancy, which results in 26 channels, readout by two SIPHRA 16-channels ASICs. The entire assembly is mechanically included in an aluminum frame.

Spectral performance evaluation

Spectral performance was assessed for the Low Energy Detector based on a preliminary design of the PHEMTO DEFPET pixels, the results obtained with the MIXS instrument and the ATHENA focal plane prototype, and a model of leakage current and electronics noise with radiation damage dose. The

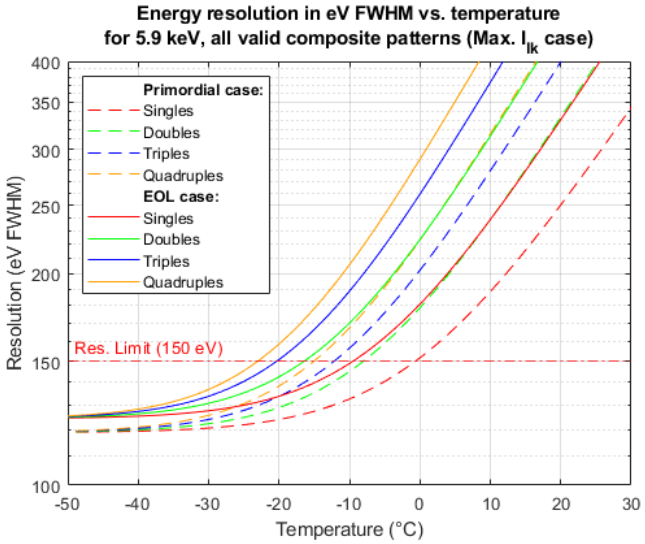


Fig. 3.7: PHEMTO/LED energy resolution at 6 keV (FWHM) at the beginning and end of life (EOL) vs detector temperature. For end of life performance, we used conservative hypotheses: 10^6 10 MeV-equivalent protons/cm^2 displacement damage dose, $250 pA/cm^2$ leakage current and 5 electrons rms readout noise.

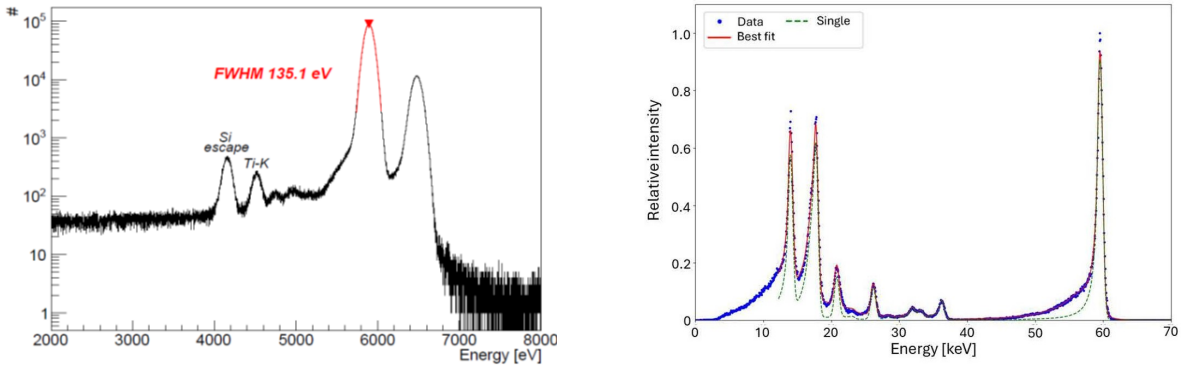


Fig. 3.8: PHEMTO spectra obtained with the LED and HED prototypes demonstrating 135 eV FWHM at 6 keV [Meidinger et al., 2017] and 0.9 keV FWHM at 60 keV [Allaire, H. et al., 2025]

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equatorial orbit of PHEMTO is optimal for non X-ray background and displacement damage dose (DDD): we estimate the 10 MeV-equivalent proton DDD after 10 years to be 10^6 protons/cm^2 in a 5° LEO behind 1 mm Aluminium shielding (very conservative value). Fig. 3.7 gives the energy resolution at 6 keV beginning of life and end of life for the various pattern multiplicities. Note that with the PHEMTO macropixels of 250 μm, we expect 95 % of the photon events to be single-hit or double-hit events. The energy resolution requirements of 150 eV FWHM can be obtained with a sensor colder than $-17°C$. The design of the trigger side is a compromise between the capacitance of the electrodes and the number of readout channels. By segmenting the cathode in 64 areas, the capacitance can be reduced to 22 pF. Using the 32-channel IDeF-X HDBD as the readout ASIC [Baudin et al., 2022], we could reach a low-energy threshold of 3.6 keV for a 10 μs peaking time, in compliance with the scientific requirement of 4 keV threshold for the coincidence with the ACD. A time resolution of 1 μs can be obtained by time walk correction in the Data Processing Unit for efficient background rejection by anticoincidence.

Spectral performance for the High Energy Detector is inferred from the MC^2-1K prototype of hybrid detector with 250 μm CdTe pixels and D_2R^2 front-end ASIC [Allaire, H. et al., 2025]: we obtained an energy resolution of 0.9 keV FWHM at 60 keV on the sum spectrum (see Fig. 3.8), in line with the scientific requirements of $E/\Delta E = 50$.

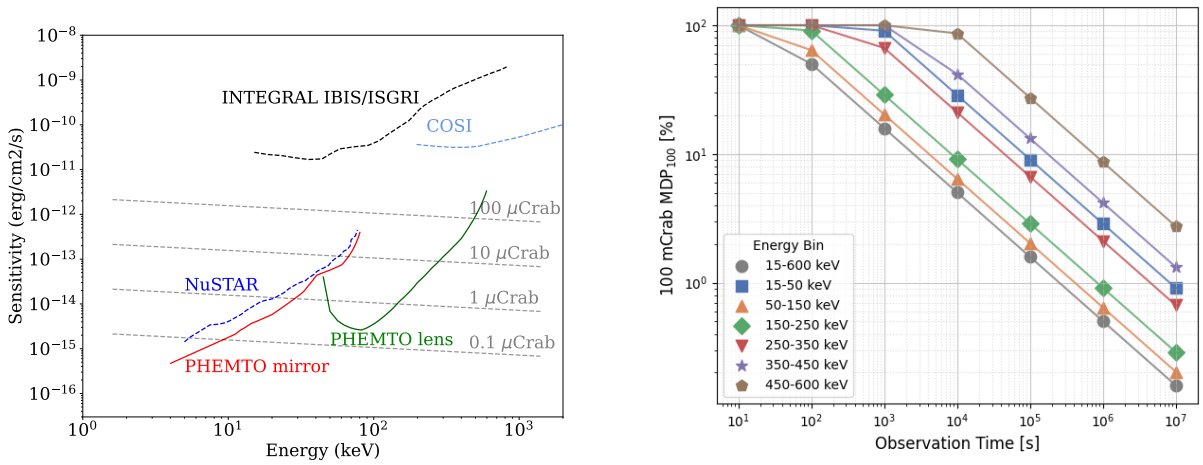


Fig. 3.9: Left: PHEMTO 3σ broadband sensitivity in erg cm^{-2} s^{-1} for 1 Ms observation, $\Delta E = E/2$. A comparison with INTEGRAL/IBIS, NuSTAR and COSI is also shown. Right: MDP for Crab-like source with a 100 mCrab flux integrated in different energy bands vs observation time (s).

Polarimetry performance evaluation

To access the polarimetric performance of the PHEMTO instrument configuration a Monte Carlo simulation study was performed using the MEGAlib toolkit, [Zoglauer et al., 2006], for the energy range 15 keV-400 keV since for $E_\gamma < 15$ keV the Compton cross section of the LED Si material is very low.

To determine the optimal FPA configuration, we studied the system polarimetric response by varying the distance between the LED and HED, as well as the HED planar size. Since detector energy resolution affects the Compton reconstruction algorithm, we assumed at this stage a noise level of 74 e$^-$ for the HED up to 200 keV, and a constant 0.5% in the range 200–400 keV. For the LED, we assumed an end-of-life energy resolution of 150 eV across the full energy range. For lower energies (<100 keV), we found that reducing the distance between detectors improves polarimetric performance, as it increases both the Compton detection efficiency and the μ_{100} modulation factor (see below). At higher energies, no significant dependence was observed on either the inter-detector distance or the HED planar size. Therefore, the HED size is driven by imaging requirements. Considering mechanical and assembly constraints, a spacing of 1 cm between the LED and HED was selected (Figure 3.6) and used for the results presented below.

To estimate the instrument capability of detecting a % of polarized photons within a population with a confidence level of 99% the Minimum Detectable Polarization (MDP) metric is often used [Weisskopf et al., 2010]. The MDP is given by:

$$MDP = \frac{4.29}{\mu_{100}\varepsilon_{Compton}N_S}\sqrt{\frac{\varepsilon_{Compton}(N_S+N_B)}{T}} \tag{3.2}$$

where μ_{100} is the modulation amplitude for a 100% polarized photon source, $\varepsilon_{Compton}$ is the Compton event identification efficiency, N_S is the source photon flux irradiation the detector system and N_B is the background rate at the PSF area. PHEMTO optical system configuration allows $N_S \gg N_B$.

In order to correct for systematic, geometry-induced modulations in the azimuthal distribution of Compton events, two photon populations were simulated: fully (100%) polarized photons and background non-polarized photons. The fundamental FPA polarization performance was studied using monochromatic and Crab-like on-axis point sources (see Figure 3.9).

3.4.3 Detector Control Electronics

The Detector Control Electronics will be composed of three independent modules (LEDE, HEDE, ADCE), either placed in three separate warm electronics boxes or grouped in a single box. A trade-off study will be done on the DCE mechanical configuration in phase A taking into account required power, thermal management, interfaces and integration constraints with the different partners. This section summarizes the main functions to implement.

LEDE functions

The Low Energy Detector Electronics (LEDE) is in charge of the LED readout. This consists in supplying and controlling the ASICs and the analog to digital conversion of the energy information, on both the pixel side (sensor frontside) and the trigger side (sensor backside). For the pixel side, it requires to implement a sequencer for a rolling shutter mode readout and a fast ADC for a 25 Msamples/s sampling rate. Flexible application of window mode upon request shall be also managed by this sequencer. For the trigger side, it requires a controller with a fast trigger detection but slow data encoding. The LEDE will be also in charge of the high voltage supply (-250 V) for sensor depletion and more generally the proper switch-on and switch-off power sequences according to the LED requirements. The LEDE will perform data pre-processing to strongly reduce the data rate at the output of the detector system. The goal is to convert frame data in hit pixels lists with time information, pixel coordinates and pixel data. This implies to implement on the fly a number of operations on the rows like pixel offset computation and correction, common mode correction, pixel threshold computation operation. The clustering of hit pixels can be done in this board or later in the pipeline.

HEDE functions

The High Energy Detector Electronics (HEDE) is in charge of the HED readout. This consists in a separate control of the 36 front-end ASICs (4 by detection unit) and the 9 embedded ADCs to extract for each trigger the information of the hit pixels and the deposited energy. The dead time will be managed independently for each ASIC. The HEDE will be also in charge of the high voltage supply on the CdTe cathode (amplitude depending on the final detector thickness, $\sim -800V$ for 2 mm) and the low voltages supplies for the detection units. Typical frequency for the control clock is 20 MHz.

ACDE functions

The Anticoincidence Detector Electronics (ACDE) is in charge of the active shielding readout. This consists in the control of the front-end ASIC to detect the trigger. For each event, the readout electronics will send the trigger timestamp, with a 50 ns accuracy, and the number of the SiPM which have triggered. This may enable a "veto zoning" where we could identify, if needed, which plastic plates have triggered. The ACDE will be also in charge of supplying the high voltage for the SiPM ($\sim 50V$) with a thermal control loop to guarantee a stable gain of the photodetection during the observation.

Detector power supplies definition

Some electronics board will be in charge of generating multiple secondary supplies for the detectors based on a primary power supplied by the Instrument Control Unit. The board will guarantee the noise performance required by the detectors with the required operability (tunable voltages). The real power distribution scheme between the detector power supply board and the LEDE, HEDE, ADCE will be studied in phase A.

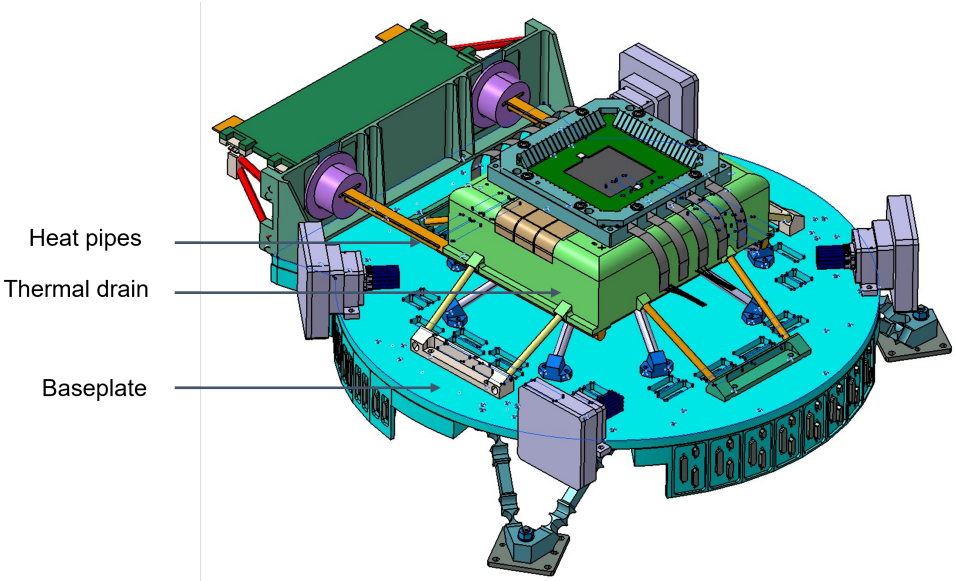


Fig. 3.10: Mechanical and thermal design of the PHEMTO focal plane assembly. The LED and HED are cooled down thanks to heat pipes and a massive part acting as a thermal drain and thermally isolated from the baseplate which is the mechanical reference for the telescope.

3.4.4 Filter Wheel Assembly

The filter wheel will implement:

- A shutter for background calibration, protection to Sun light if necessary
- An open position for standard observation
- A UV/visible filter to reduce fluxes of bright sources in UV/optical in the field of view.
- A calibration position with a radioactive source to be defined (for instance ^{133}Ba like in Solar Orbiter/STIX with spectral lines at 4.7, 31, 35, 81, 276, 302, 356 keV to calibrate both Si and CdTe over the full energy range).

3.4.5 Collimator structure

To reduce the cosmic X-ray background component for lower energies on the PSF area of the detector system, we will use a 0.13 cm thick lead (Pb) cylindrical collimator positioned 5 cm above the LED. Taking into consideration the lens size, viewed at the focal length of 20 m we have determined its apperture angle to be $< 5°$, implying a collimator length around 100 cm and an inner radius of 7.6 cm. This configuration will allow >99 % X-ray diffuse background suppression at the PSF area for energies until ≈150 keV.

3.4.6 Thermal control system

The thermal control system will be based on passive cooling with large radiators and low conductance heat pipes going through the focal plane mechanics (see Fig. 3.10). In the HED layer, the thermal gradient between the connection pins of the detection unit and the sensing part (CdTe) will stay below $15°C$ thanks to low power ASIC (0.3 mW per channel, 2.8 W per detection unit). In the LED layer, the sensing part (Si DEFPET) will be thermally decoupled from the front-end ASIC and thermally coupled to the thermal drain. A preliminary thermal study proposes a $1m^2$ radiator that could be maintained below $\sim -45°C$ to provide the required temperatures for the detectors ($\sim -30°C$ for Si, $\sim -15°C$ for CdTe). A second radiator will be necessary to the DCE but could be smaller ($0.5m^2$) and much warmer ($\sim 10°C$) to operate the electronics boards close to room temperature. The baseplate will be the mechanical reference for the alignment of the FPA with the telescope and also for the mounting of the detector electronics boxes opposite to the FPA.

3.4.7 Key technologies and heritage

ASIC technologies

Front-end ASIC for the LED will be VERITAS-2 and SWITCHER for the pixel size, in the same versions as those designed for the ATHENA mission. The VERITAS-2 is a 64-channel combined amplifier-shaper circuit featuring trapezoidal shaping for optimum series noise suppression, with integrated shaping sequencer and serial analog readout. It is made in the AMS 0.35 μm 3.3 V CMOS technology. The current version is V 2.3 features a fully differential architecture and provides multiple test and probe features. It supports both commonly used DEPFET readout modes, drain current readout and source follower readout. For the LED, the drain current readout would be the preferred option, as in this mode, the readout node capacitance of the large detector does not slow down the readout. A precursor chip, the so-called ASTEROID, which is used as front-end amplifier for MIXS on board of the BepiColombo spacecraft. As the VERITAS II has some differences in architecture, a requalification is done in the framework of the ATHENA programme. The Switcher-A is a 64-channel, 3-port analog HV switching circuit specifically designed for the readout of large DEPFET arrays. It is required for switching the Gate, Cleargate and Clear terminals of the DEPFET rows. Its feature set includes fast switching (rise times < 100 ns depending on load capacitance), a power efficient, latch-driven switch/hold current toggle with one row prefetch, a round robin type token passover mechanism for daisy-chained operation of several ICs, a fast shift-register controlled row selection mechanism for random access of rows and an open-drain, daisy chained heartbeat mechanism for SEU detection, monitoring of health and control of synchronisation. Like its precursor, the Switcher-S, which was used for MIXS on BepiColombo, this chip is designed in the AMS 0.35 μm HV CMOS technology and designed by the STZ (Steinbeis-Transferzentrum Microelectronics and Sensor systems) under the lead of Prof. P. Fischer. The precursor IC, Switcher-S, was fully tested to MIXS irradiation levels for the use with the MIXS detectors on board the BepiColombo spacecraft. The Switcher A is constructed identically, with the exception that for efficiency reasons each IC has 3 instead of 2 HV output ports per channel. Testing to ATHENA irradiation levels in in progress.

ASIC solutions for the LED trigger side will be investigated during phase A. The current baseline is IDeF-X HDBD designed by CEA-Irfu in AMS 0.35 µm CMOS technology as low noise radiation hard chip with 32 channels and a global trigger logic [Baudin et al., 2022]. Its design is very similar to IDeF-X HD (difference in the input stage only) flying since 2020 on-board Solar Orbiter in 32 CdTe spectrometers with no drift in performance. ASIC solutions for the HED detector will be the sucessor of D2R2 and ANAQINX designed by CEA-Irfu with XFAB CMOS 0.18 µm technology [Boulfani, H. et al., 2025]. These 2 ASICs have been developed in the framework of PhD half founded by CNES, to address both performance and space quality. The challenge is to integrate in a surface not larger than 250 µm x 250 µm a low power spectroscopic channel. In 2025 we demonstrated the low noise performance (<25 electrons rms, 0.3 mW/channel) even after 30 krad total ionising dose.

ASIC solutions for the ADC SiPM will be investigated during phase A. The current baseline is the SIPHRA ASIC from IDEAS company as radiation hard chip with flight heritage on board, for instance, the Eirsat-1 Cubesat.

DEPFET detector technology

Design and fabrication of the DEPFET based LED detector will be done at the MPG seminconductor laboratory (MPG Halbleiterlabor, HLL) in Garching in the proprietary HLL DEPFET technology. Apart from minor optimizations the technology will be the same as used for the Focal Plane detectors of MIXS on BepiColombo [Majewski et al., 2012], and for the ATHENA Wide Field Imager. In the scope of PHEMTO, the ATHENA pixel structure, which is superior in terms of noise and clear behaviour, is placed into a driftrings structure similar to that of BepiColombo, with the driftrings being slightly reduced in size to match the pixel pitch requirement. The effect of NIEL and annealing on the detectors has been

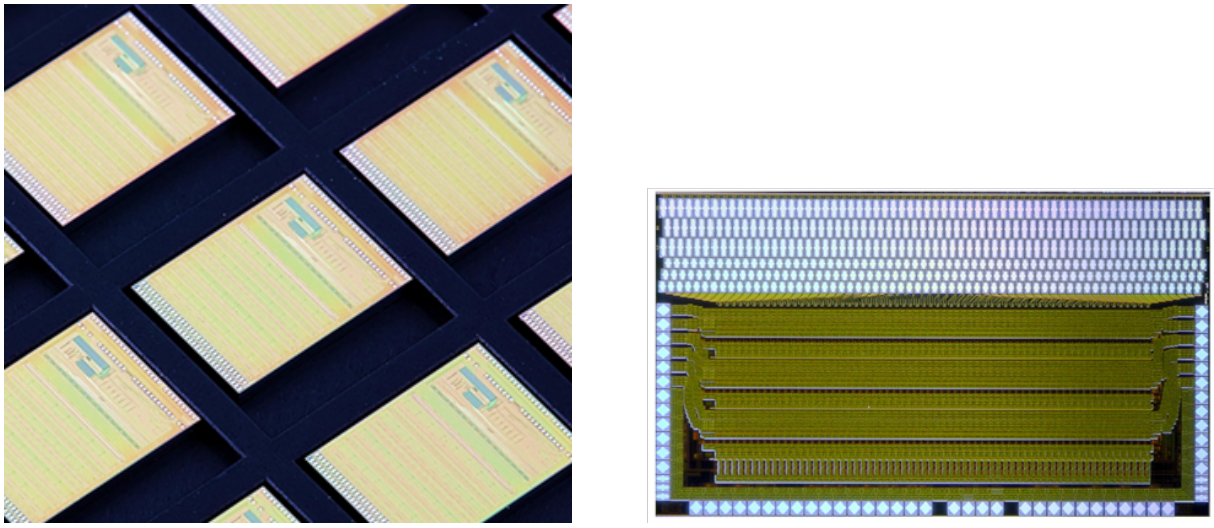


Fig. 3.11: Integrated circuits for the readout of the LED (Left) Veritas II (Right) Switcher-A

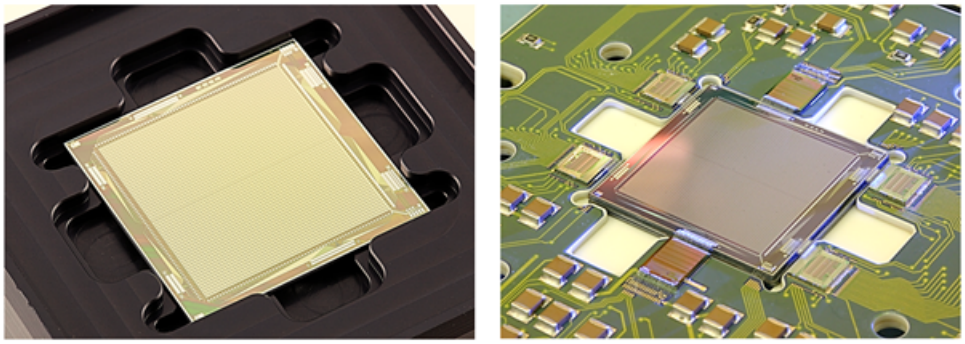


Fig. 3.12: Macropixel detectors for MIXS in carrier (left) and bonded on prototype hybrid (right).

studied in the scope of BepiColombo, as here the impact of bulk damage is dominant due to the high equivalent flux and a relatively high detector temperature was required. The tradeoff between TID and detector resolution is under study for ATHENA.The performance models for the PHEMTO LED detector are based on these measurements.

CdTe hybrid detector technology

The technological challenges for the PHEMTO CdTe hybrid detectors are on one hand, to provide time-resolved imaging and spectroscopy with a spatial resolution of 250 µm and on the other hand, to cover a detector area larger than 30 cm^2. CEA-Irfu designed a 2D ASIC to be point to point connected to a pixel CdTe detector with the required pixel pitch. A prototype of 32×32 pixels showed an energy resolution of 0.9 keV FWHM with a really good pixel-to-pixel response uniformity and few dead pixels (< 0.5 %) [Allaire, H. et al., 2025]. To cover a large area, CEA-Irfu has worked for 20 years with the industrial partner 3D PLUS to develop space-qualified packaging technologies for sensor technologies. They designed Caliste-SO spectrometers successfully operated since 2020 in Solar Orbiter with excellent spectral performance, based on System-in-Package (SiP) technologies. For PHEMTO, the goal is to use the more recent Wireless Die-on-Die (WDoD) technologies to form mosaics of ASICs with minimum space in-between and to connect all the ASICs signals underneath the sensor area with a pitch of 250 µm. Then the SiP technologies will be also used to implement an ADC chip and passive parts into a compact detection unit to mount on a printed circuit boards like for Solar Orbiter. Prototype of 64×64 pixels with CdTe, front-end and ADC is under design for a production in 2027, in the framework of a R&T

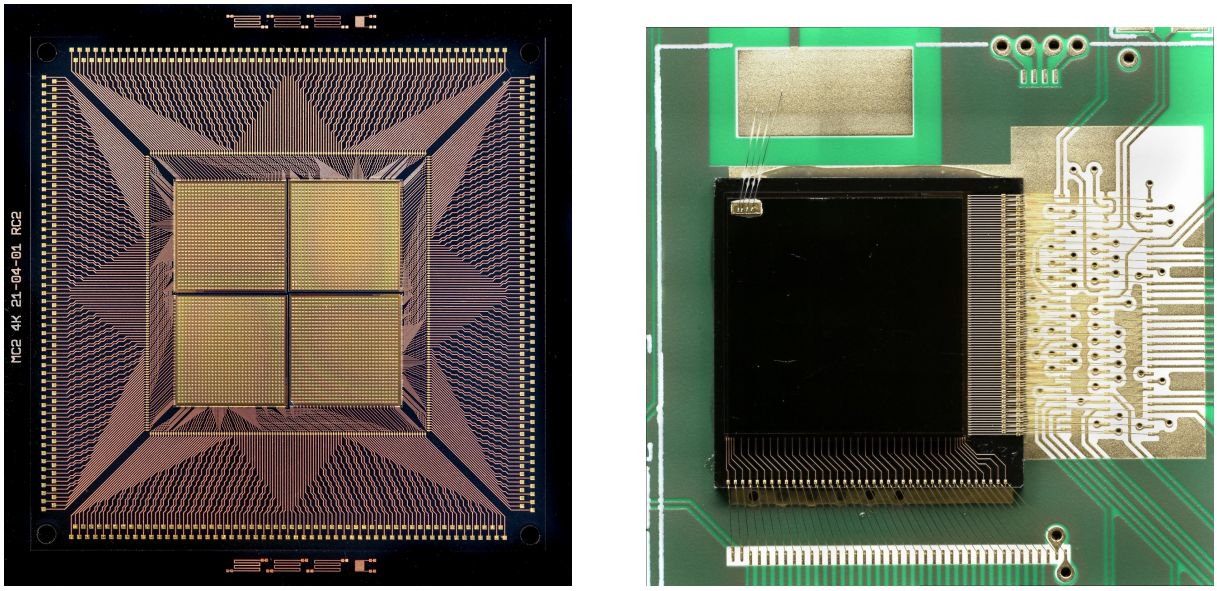


Fig. 3.13: (Left) Mosaic of front-end ASICs in a WDoD layer forming an array of 64×64 pixels. (Right) Prototype of CdTe hybrid detector with 32×32 pixels.

program with the French space agency. The PHEMTO detection unit will require a redesign of the ASIC from 32×32 to 48×48 pixels to realise a 96×96 pixel detection unit. This allow a configuration on the detection plane with a central detection unit of 18 mm$\times$ 18 mm active area, for good polarimetric measurements, even if the PSF is moving in the focal plane due to a drift of the line of sight of the telescope. The 9×9 unit configuration limits the dead zones in the focal plane to perform high performance imaging of extended sources.

Filter wheel design

The team at the University of Geneva has an extensive heritage in the development and provision of mechanisms for space and filter wheels, in particular. The team has developed in collaboration with several Swiss industries the filter wheel that was launched on-board Hitomi (2013), the filter wheel on XRISM (2023), and has recently (2021) provided the RSU shutter unit which is being operated on the VIS instrument on-board the ESA-led Euclid mission. The same team is currently responsible for the filter wheel mechanism of the X-IFU instrument on NewAthena, the filter wheel of the IRT instrument on-board THESEUS, and is currently in the process of beginning the prototyping phase of the UVIM door mechanism. The latter is to be mounted on the NASA-led UVEX mission and is planned for a launch in 2030.

The design of the PHEMTO filter wheel could be based from the wheel of Microchannel X-ray Telescope of the SVOM mission launched on 2024 with a Sagem motor, a pinion and a cog wheel (see Figure 3.14). This mechanism was qualified for 150000 rotation cycles due to regular passages in the South Atlantic Anomaly. This mechanism in PHEMTO is not operated at cryogenic temperatures and will require less cycles in equatorial orbit. It does not require specific development and qualification (other than standard mechanism qualification in the ECSS).

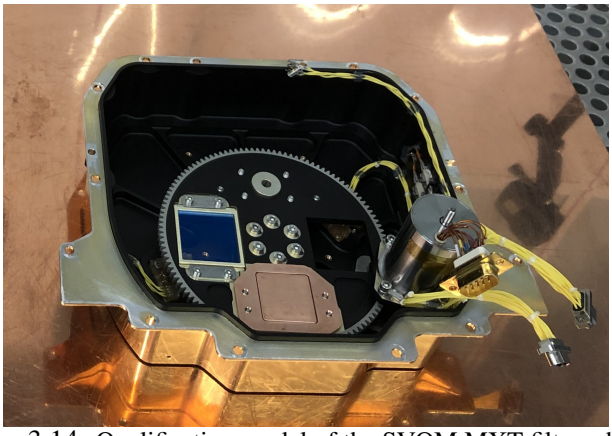


Fig. 3.14: Qualification model of the SVOM MXT filter wheel

3.5 Telescope system

3.5.1 Design principles

The telescope system is a key element of the instrument to maintain the alignment of the mirror system with a focal plane array placed 20 meters away. A pointing stability of the order of 1 arcmin could be acceptable if the pointing knowledge error is better that the point spread function, taking into account the thermo-elastic deformations of the telescope. This is why the telescope will be associated with a alignment system based on laser to correct for the instantaneous position of the line of sight. The time-tagged information will be used to correct images and polarimetry data on ground. This post-processing is possible by downloading the raw content of the time-tagged photon event lists, which is highly compatible with low-Earth orbit telemetry capability in X band. Another technical challenge is the deployment of the 20-meter mast folded in the rocket fairing.

3.5.2 Key technologies and heritage

Extendable boom

The boom technology, mostly developed in the USA and in Japan, starts being available in Europe and is a truly accessible goal for a mission in 2041. The identified provider for the PHEMTO mission is the Active Space Technology company in Portugal (AST, https://www.activespacetech.com/). AST is developing and qualifying the MAIT process of light long tubes made of Ti6AL4V, being a low CTE material with the best ratio of stiffness and mass that metals can offer, in the framework of the ESA Envision mission. AST has recently successfully validated a BBM of a custom rolling machine of Ti6AL4V sheet, which is one of the greatest challenge on this MAIT process.

The PHEMTO boom will be derived from the 16 m extended boom developed for the ENVISION/SRS instrument. Detailed mechanical and thermal studies of this boom will performed on the next phases in order to get a detailed design. If we extrapolate from the SRS boom, we can expect a boom mass of the order of 20 kg. Currently, AST is finalizing samples and tooling design and some de-risking was already performed at BBM level in order to ensure a smoother MAIT process. Thus, the qualification process is on-going in order to reach TRL 6 before 2030.

Telescope alignment system

The telescope alignement system will enable to measure and reconstruct the telescope line of sight (LOS). A pointing stability of the order of 1 arcminute is acceptable, if this system enable us to measure with arcsecond precision the LOS typically every seconds. A very performant 3D system (CAMS) was developed by the Canadian Space Agency, based upon lasers, for the HXI instrument oboard the Hitomi Japanese mission. The HXI was at the end of a boom 8 meters away from its X-ray mirror, and

the thermo-mechanical boom and LOS modification was measured with arcsecond precision. Such a technique is thus completely mature. In Europe, the Osservatorio Astrofisico di Torino (INAF) has been contacted in order to provide a similar alignment system.

Telescope requirements

We propose that ESA takes the lead in the system requirements and the procurement of this telescope mechanisms (boom and alignement system) by managing the European industry. ESA large feedback experience of missions with pointing constraints would be very valuable for this ambitious telescope.

3.6 Instrument Control Unit

The instrument control unit, mounted on the spacecraft, will include two processing boards and two power supply boards working each in cold redundancy. The power supply boards will mainly provide independent primary powers to the detector system, the telescope system and the optics system. The amplitudes will be defined according to the electrical architecture and power requirements of each assembly. Possibly it will unfiltered powers from the platform (28V) and a switch to supply or not the assembly. If this implies too much power dissipation close to critical items (detectors, optics, lasers), intermediate amplitudes with DC-DC converters will be implemented in the ICU. PHEMTO electrical architecture is shown on Figure 3.15.
The processing boards will receive, process, store and send via SpaceWire (SpW) telemetry the science data, look after and reports on health and Instrument Fault Detection, Isolation, and Recovery (Instrument-FDIR) and run the flight software. The ICU communicates with the spacecraft by a standard interface, e.g., SpaceWire or SpaceWire and 1553, with time provided as SpaceWire time-codes or a pair of separate (redundant) one-pulse-per-second (1PPS) interfaces.

The ICU receives the data from the detectors via pairs of (redundant) SpaceWire links. It then sorts the data types into different streams (e.g., science, housekeeping) and processes them based on coincidence. Events in time coincidence between LED and ACD or between HED and ACD will be flagged and discarded from the telemetry in observation mode keeping the statistical information. Events in time coincidence between LED and HED will be flagged as potential Compton events and candidates for polarisation measurements. Time coincidence window will be typically 1 μs.

Other data processing close to the detector system will be defined in a more accurate way in phase A. This can include science quick look products (light curves, spectra, images), noisy pixel management, photon position correction after reconstruction of the line of sight by the telescope alignment system.

On the ICU's CPU-board is a Frontgrade Gaisler GR740, boots a golden boot image from a read-only MRAM and runs application software from a read/write MRAM, allowing for software updates. The CPU uses SRAM as operating memory. The CPU board also provides the watchdog and sends the detector electronics accurate 1PPS instrument time.

The ICU also takes analogue HK measurements of the ICU box provides and switches 28V power to the detectors. The flight software runs on the RTEMS operating system and is developed on Linux avoiding proprietary and closed-source software and drivers to maximise maintenance possibilities during the long development and operation duration. The software running on the CPU provides the following:

- Telecommanding
- Telemetry (standard ESA PUS telecommand verification, housekeeping, events, science data, etc.)
- Control and communication with the detectors
- Internal instrument limit checking
- FDIR
- Power control
- Gathering health data from the ICU and detector electronics
- Memory maintenance

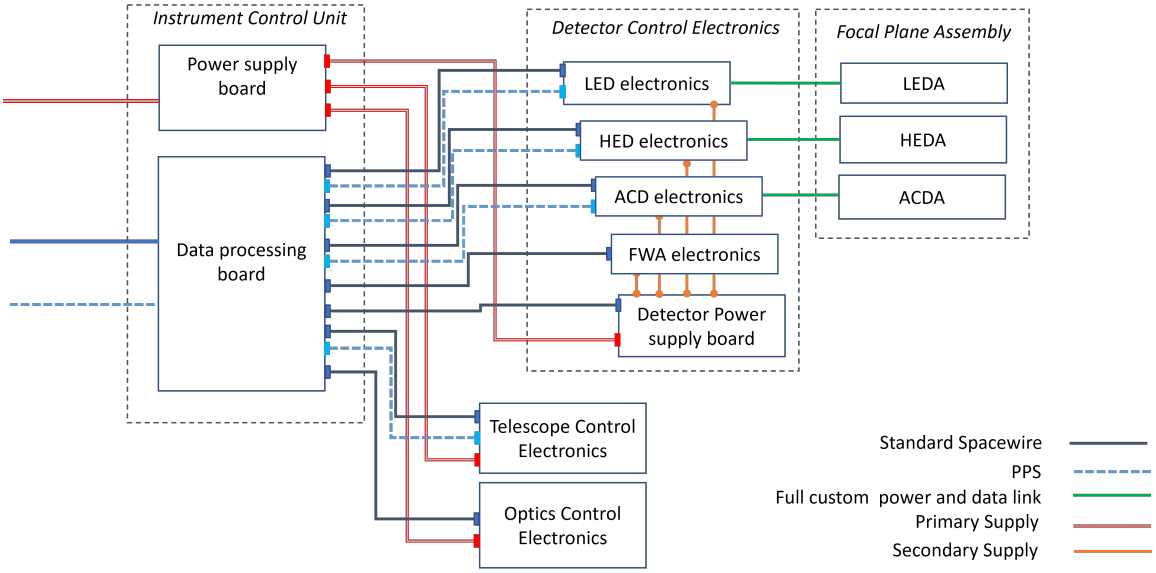


Fig. 3.15: PHEMTO electrical architecture. Redundancy is foreseen but not shown here, not to saturate the diagram.

- Timekeeping and synchronisation
- Watchdog control
- Calibration
- Science sequencing
- Science data compression

The ICU will be developed with simulators (camera simulators, spacecraft simulators) which will be built as simply as possible and avoiding closed-source software and drivers, with as long a life as possible to allow the project to meet the development and long-term maintenance requirements. Software PA will be provided by the software group overseen by the PA group.

3.7 Required Resources

Estimated mass includes a total optics mass of 530 kg (mirror: 290 kg and lens: 240 kg), including a 20% margin. These numbers are derived from the mirror design and the last LAUE lens R&D studies. Studies for the COSPIX mission lead us to a detector payload around 170 kg with margins, which should be similar for PHEMTO. If we scale it to the one already realized for ENVISION, the boom mass should around 50 kg, margins included. Conservatively assuming service module of around 630 kg for a total dry mass around 1400 kg. Estimating 300 kg for the adapter and propellant for the satellite gives a total wet mass of 1700 kg, well within the constraints of the Ariane 6.2 launcher. The peak payload power requirement is estimated to be around 500 W, and the envisaged scientific data rate is around 2 Mbps.

3.8 Specific Interface Requirements

3.8.1 Attitude and Pointing System

The required absolute pointing reconstruction is 3 arcsec at the 90% confidence level (SR11) and the angular resolution (HEW) should be around 10” depending on energy (SR 6). These pointing reconstruction and angular resolution requirements imply constraints on the alignment and pointing of the optics. System studies made for the COSPIX M3 mission, adapted to PHEMTO, have shown that the mirror absolute pointing accuracy should be then around 20" to meet these requirements. Attitude reconstruction should have also arsecond accuracy in order to identify precisely the sources in the PHEMTO field of

view and meet SR-11 requirements. A full telescope system study will be made in later phases under ESA responsability.

3.8.2 Telemetry

The sources observed by PHEMTO are different in nature and intensity. Moreover, a lot of them are intrinsically time-varying. The amount of scientific data generated between two download periods will thus be variable and enough data storage should be foreseen onboard. Most of the observations will be on low brightness sources, generating a low amount of data. The communication requirements with the ground will be thus driven by the less frequent observations of very strong sources. The data corresponding to such strong sources (intensity of several Crab) that will be observed for a relatively short time (typically 10 ks or less) can be downloaded in a few standard contact sessions. It is estimated that these conditions are met with the following requirements:

- Scientific data volume : 3 GBytes/day
- Telemetry rate : 2 Mbits/s

In order to be able to continue nominal operations with ground contact for a period of 7 days, an internal storage of $\approx$ 30 Gbytes is needed, which is well within the capacities of present on-board mass memory storage.

3.9 Calibration

The objective of a calibration is to determine a response function which maps a known input function into a precisely measured output function. Knowledge of the calibrated response function will then allow use of output data (on-orbit observations) to deduce the input data (X and gamma-ray properties of cosmic sources).

As a starting point the team will create a simulated detector responses that will be verified against calibration measurements at detectors, instruments and Payload levels. System-level calibration using the FM electronics will be performed after the instruments are fully assembled. Imaging simulations of the full telescope will be also done using the SIXTE software [Dauser et al., 2019], developed at ECAP/Remeis Observatory, in Germany.

Detectors calibrations will be performed using various different sealed radioactive sources producing line emission at various X and Gamma ray energies with the scope to asses: energy response, angular response, timing response, trigger threshold of the detectors. The full detector payload polarization response will be calibrated in a X-ray synchrotron accelerator, such as the ESRF in Grenoble (France).

The mirror module performance will be verified, in addition to metrology/reflectometry tests performed by the manufacturer on mandrels and mirror samples, via pencil-beam tests at PANTER by means of the 20-m focal length extension developed in 2012 for the IXO project (Burwitz et al. [2013]). This will enable the reconstruction of the real effective area and the point spread function of the optical module. The lens will be tested at the different stages of its realization at the LARIX facility in Ferrara (Italy).

In flight imaging and spectral calibration will be defined as well using standard candles and intense cosmic sources. The energy response of the detectors will be also monitored making use of the filter wheel radioactive sources. The timing performances will be verified using pulsar observations. The resulting calibration files will be used to make end-to-end simulations with SIXTE and to perform the in-flight data analysis (see section 4.2.

4. Mission Configuration and Profile

The mission profile is driven by the following top level requirements:

- PHEMTO should be operated in a pointing mode, with observation times which can range from a few kiloseconds to several hundreds of kiloseconds. In addition, the full sky should be accessible at least once during the mission.
- PHEMTO is an observatory, with an observation program that will be regularly uploaded according to scientific selections, and mission constraints. As a lot of sources are variable in time, it is highly desirable for variability studies of source to minimize possible interruptions during an observation sequence and adopt a flexible profile for the observation plan. However, no fast repointing after TOO requests is required.
- There is no scientific necessity, during routine observations, to have a continuous link with the Earth. The spacecraft can be operated autonomously, following an observation program that will be regularly uploaded.
- The full scientific data should be downloaded regularly at least for monitoring the telescope behavior at regular time intervals.

4.1 Orbit Requirements

The prefered orbit for PHEMTO is an equatorial Low Earth orbit (LEO) with an inclination less than 5°. This LEO orbit provides the best background environment for a high energy telescope as it avoids South Atlantic Anomaly passages. Thus, the expected background will be low, compliant with SR-9, and the particle environment will be quite safe for the different payload and spacecraft electronics. Indeed, we estimate the 10 MeV-equivalent proton dose after 10 years to be 10^6 protons/cm^2 behind 1 mm aluminium shielding, which is much lower than what is expected for more inclined or polar LEO and higher altitude orbits. This will also enable us to keep a good energy resolution for the two detector systems during the whole mission duration, even after an extension period.

4.2 Ground Segment and Communication Requirements

The Ground Segment (GS) organization we propose in this section provides the means and resources to manage and control the mission via telecommands, to receive and process the telemetry from the satellite, and to disseminate and archive the generated products. The overal PHEMTO GS organization is proposed following the usual standard assumptions for similarly sized-missions led by ESA. The Operations Ground Segment (OGS) consists of the Mission Operations Centre (MOC) and the ground stations which receive the telemetry. The Science Ground Segment (SGS) consists of the Science Operations Centre (SOC), the Science Data Centre (SDC) and the Instrument Operations Centre (IOC). The development and operations of the SDC and IOC are nationally funded by the member states participating to the project. An overview of the GS is given in Fig.4.1. All components of the OGS and SGS are described below, while a brief description of the planned PHEMTO data levels (0 to 3) is provided in Sect. 4.2.1.

- **Mission Operations Centre (MOC).** The MOC is the only interface with the spacecraft. It will be based in Darmstadt (Germany) and held responsible for the commanding of the spacecraft and instruments (including the overall mission planning), for ensuring the spacecraft safety and health, and for provision of Flight Dynamics support. The MOC performs all communications with the satellite through the ground stations for the upload of the platform and payload telecommands (based on the observation schedule provided by the SOC, see below), and reception of the downloaded telemetry data. They are also responsible for collecting the science data and its transmission to the SOC, along with the raw telemetry, housekeeping and auxiliary data. It is assumed that the main ground station used for PHEMTO is Malindi. Due to the non-continuous coverage, the spacecraft will be mainly controlled via off-line operations when not in contact with the ground station. After

the initial spacecraft commissioning, all telecommands required to carry out the mission will normally be loaded in advance on the Mission Timeline for later execution. The spacecraft will be able to continue nominal operations without ground contact for a period of up to 7 days. On-board Control Procedures will allow autonomous execution of complex procedures, including decision loops which the GS cannot support due to the limited ground coverage. The MOC will provide telecommand history and other auxiliary data to the SOC.

- **Ground stations.** The science data rate estimated for all instruments on-board PHEMTO can be managed through a standard number of 10-15 daily passage over Malindi assuming an X-band. A potential additional ground stations option is the one operated by ESA in Kourou. The estimate about the telemetry required by PHEMTO has been based on the assumption that the sources observed along the mission lifetime are different in nature and intensity, with the bulk of these objects being also intrinsically time-varying. It is estimated that the telemetry volume per day will be about 3 GBytes/day (including house keeping and all scientific data streams).
- **Science Operations Centre (SOC).** The SOC will be based at the European Space Astronomy Centre (ESAC, Spain). It will be the single point of contact to the MOC for providing detailed operational requests, and it will plan all payload operation activities. The SOC is also the first point of contact of the scientific user community (through a centralized helpdesk), taking care of the yearly release of Announcements of Opportunities (AOs) for the community to submit their observational requests to the observatory. The SOC provides support in the exploitation of the mission throughout all mission phases. The overall SOC activities include the following:
  - Science observations planning and proposal handling.
  - Establish the coding standards and the overall environment for the development of all PHEMTO related software.
  - Processing of the raw telemetry down to the level 1 data, which is the level that is easily manageable by scientific community using publicly distributed software (the data processing at SOC will take place by using all necessary calibrations provided by the IOC and the software provided by the SDC).
  - Set-up and maintenance of the PHEMTO Science Data Archive (SDA).
  - Run the observation and performance simulator which is built in collaboration with the IOC.
  - Set-up and maintenance of the centralized help desk to provide support to the scientific community with respect to any PHEMTO related issue (instruments, data analysis, scientific exploitation).
  - Public outreach and organization of workshops on the PHEMTO data analysis and mission products exploitation.

  It is assumed above that the lowest level processing of the PHEMTO data is carried out at SOC, in order to guarantee at any time, the availability of fully usable data to the science community. A standard set of higher level products can be produced for each observation by the SDC and made available to the science community through the SDA. All the PHEMTO data processing software is developed by the SDC in collaboration with the IOCs. However, part of this software will be provided and installed at SOC for local execution during the lowest level processing. The ESA PHEMTO Project Scientist will be part of the SOC team. It is expected that all decisions concerning deviations from the nominal PHEMTO observing plan reside within the Project Scientist.
- **Science Data Centre (SDC)** The SDC will be established through national funding and will be led by Switzerland, with support from most of the member states joining PHEMTO. Following the example of past ESA-led missions and missions being planned for the future in the field of high energy Astrophysics, the SDC is thus organized as a consortium of institutes led by a centralized entity which ensure a smooth functioning and provides a single interface to the Agency (University of Geneva, Switzerland). The SDC is mainly responsible for the development of all PHEMTO related software. The latter is needed both for the daily pipeline data processing (from the raw

telemetry to the higher scientific level products, i.e. level 1, 2, and 3) and for the interactive processing carried out by the final scientific users. The SDC will also provide the Quick-Look Analysis (QLA) software system that is needed to inspect the instrument data as soon as they are available on the ground in order to promptly plan possible follow-up activities (if needed) and to monitor the instrument performances or carry out the trend analysis. The SDC will perform part of the daily pipeline processing of the data, accessing the level 1 data from the SDA (automatically processed there by the SOC) and producing there the derived level 2 and 3 data for dissemination to the science community. The level 2 data produced by the SDC will be used by the SDC for quick look activities but also by the IOC to spot eventual instruments anomalies and to perform systematic instruments trend analysis and performance monitoring. The quick look activities are expected to generate the dissemination from the SDC to the world-wide community of rapid communications of events which are relevant from an Astrophysical perspectives. These can include, but are not limited to the outbursts of new/known variable sources, the discovery of new celestial objects, the follow-up of sources which were discovered by external facilities operating at the same time as PHEMTO and triggering dedicated PHEMTO observations. These rapid communications will likely use channels as the Atronomer Telegrams (Atels) and/or the Gamma-ray Coordination Networks (GCNs). The SDC, supported by the IOC, will also scientifically validate all data products before distribution to the science community. The SDC will also provide support to the SOC for the organization of workshops for training of scientists in data analysis and software usage. Also, in collaboration with the IOC and the SOC, the SDC contributes to the testing and validation of new releases of the archive products and the overall SGS operational system.

- **Instrument Operations Centre.** The Instrument Operations Centre, is the place where all instrument specialists are located and where the expertize on the PHEMTO payload reside. The IOC provide support to ESA for integration of the instruments before launch, maintenance of the instruments after launch, operations, processing algorithms, calibration and long-term trend analysis of the instruments. They provide inputs for the instrument modelling at the SOC, and instrument specific data-analysis software for pipeline and interactive analysis, as well as the QLA, to the SDC. Personnel at the IOC will be also responsible for providing the on-board instrument software and its maintenance during all mission phases. During the course of the mission, dedicated IOC personnel will transfer the required instrument knowledge to the SOC, ensuring ESA capabilities to maintain the usability of the PHEMTO data product on the long term, even beyond the nominal mission lifetime and mission extension.

4.2.1 Data levels

For PHEMTO, we considered above a relatively standard definition of data levels, inspired by the consortium expertize gained in past high energy missions and other missions in similar fields being planned for the future. We usually define:

- Raw telemetry (TM): this is the telemetry stream received from the spacecraft and downlinked through the PHEMTO ground station. It comprises the science telemetry and the preliminary auxiliary files generated on-board.
- Level 0 data (L0): these data are obtained from the TM that is de-commutated and split into functionally independent parallel streams per instrument. L0 data will most likely be in FITS format, such that they can be conveniently handled by any scientist in the community through the PHEMTO distributed software, as well as other standard tools. On-board calibrations for each instrument are applied to L0 data.
- Level 1 data (L1): these data are obtained once all corrections (such as aspect correction, time calibration, barycentric corrections) and instrument-specific calibrations (such as detector gains and good-timing information) are applied to level 0 data. The L1 data can thus be considered as cleaned event files.

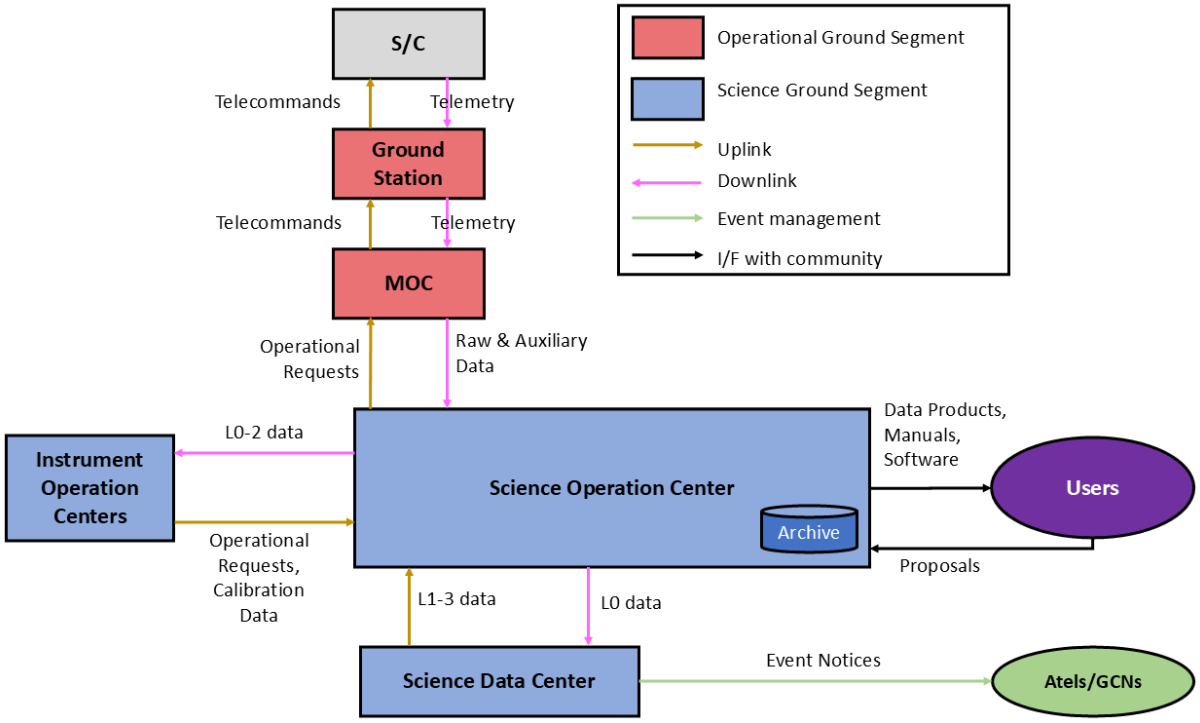


Fig. 4.1: PHEMTO ground segment (GS) organization. The GS comprises an Operational Ground Segment (OGS), under full ESA responsibility and a Science Ground Segment (SGS), which is a shared responsibility between ESA and the PHEMTO consortium. The different components of the OGS and SGS are described in Sect. 4.2.

- Level 2 data (L2): these are obtained from L1 data by applying pipelines and algorithms leading to the extraction of a set of standard scientific products per observed source and/or per observation. L2 data are provided to different partners to perform quick-look analysis activities, as well as to the community for quick inspection of the outcomes of an observation.
- Level 3 data (L3): these data are obtained from L2 data by applying higher level processing pipelines. At these stage, L3 data are not yet fully defined, but in the usual approach they are defined as longer term products that can enhance the scientific return of the mission. They can include historical measurements of the properties of the celestial sources over the mission lifetime, catalogues of the detected sources, large scale mosaics of the high energy sky, as well as cross-matches and associations with sources in catalogues derived from different energy domains.

4.3 Operational Modes

For all observations, science and in-flight calibration, PHEMTO will work in an unique pointing mode. No peculiar operation is foreseen during observations and, in nominal conditions, PHEMTO will run autonomously the observational program stored onboard. PHEMTO will have standard emergency modes in case of problems.

4.4 Mission Lifetime

In agreement with SR-12, the PHEMTO mission lifetime should be five years, commissioning and calibration phase included with a possible three years extension. The radiation environment will be computed for a ten years mission duration in an equatorial LEO. At the end of the mission, the satellite will be desorbited, in agreement with space legislation.

5. Management Scheme

PHEMTO is a common effort of European scientists experts in the soft and hard X-ray astrophysics. The present Core team is a collaboration combining institutes with experience in developing, flying, operating and managing successful gamma and X-ray science missions and with large range of both scientific and technical expertise. The involved Institutes have a well defined and precise area of responsibility as heritage of their previous activity and capabilities.

PHEMTO will be an ESA led Mission and as such an observatory facility. ESA will be responsible for the spacecraft procurement, integration of the payload, spacecraft testing and operation, acquisition and distribution of data as well as the Science Operation Centre, as described in 4.2. The scientific instruments will be provided by the PHEMTO Principal Investigator supported by the Management Board. The responsibilities may change in subsequent phases in accordance with the institute heritage and technical expertise available.

The P.I. Institute is responsible for the P.I. and overall Payload activities, with a Project Manager, System Engineer, Product Assurance Enginner. The P.I. is responsible for the design and development of the Payload as well as the scientific leadership and will review the key documentation. The Project Manager would ensure continuous monitoring of the time schedule, progresses, resources, risk, organize program meetings, support integration, testing and procedures, regularly report to the P.I.

The consortium will be funded by National Agencies and lead-CoIs will be responsible for securing funding and providing an agreed large contribution, which will be supported by a multilateral agreement between the relevant national agencies and ESA. The different entities which will supervise the mission development are the followings:

- Project Team, attended by: PI (chair), coPI, Project Manager, System engineer, QA manager, Operations Manager, S/W Manager, Payload Test Manager, Calibration Scientist.
- Management Board, attended by the PI (chair), coPI, Project Manager and the lead Co-Is.
- Calibration and Modelling Team , attended by: Calibration scientist (chair), S/W development teams.
- Data Analysis Team, attended by: DAWG chair, S/W Manager, S/W development teams
- Science team attended by PI, coPI, lead co-Is and project scientists, plus a small number of members of the wider scientific community.

The PHEMTO management board will be responsible for the different payload part procurement and the PHEMTO smooth realisation during the different phase of the project. It will meet on regular basis and could intervene in case of strong problem. The science team will discuss and optimize the scientific return of the mission during its realisation and during the mission lifetime in orbit.

6. Institute Role and heritage

The different countries contribution to the PHEMTO development is given in Figure 6.1. The different institute contributing to PHEMTO and their heritage are the following:

- Brera observatorio and Media Lario (Italy) are expert in X-ray mirrors provision and have already a unique heritage in Europe in providing mirrors for several missions from XMM to NewAthena.

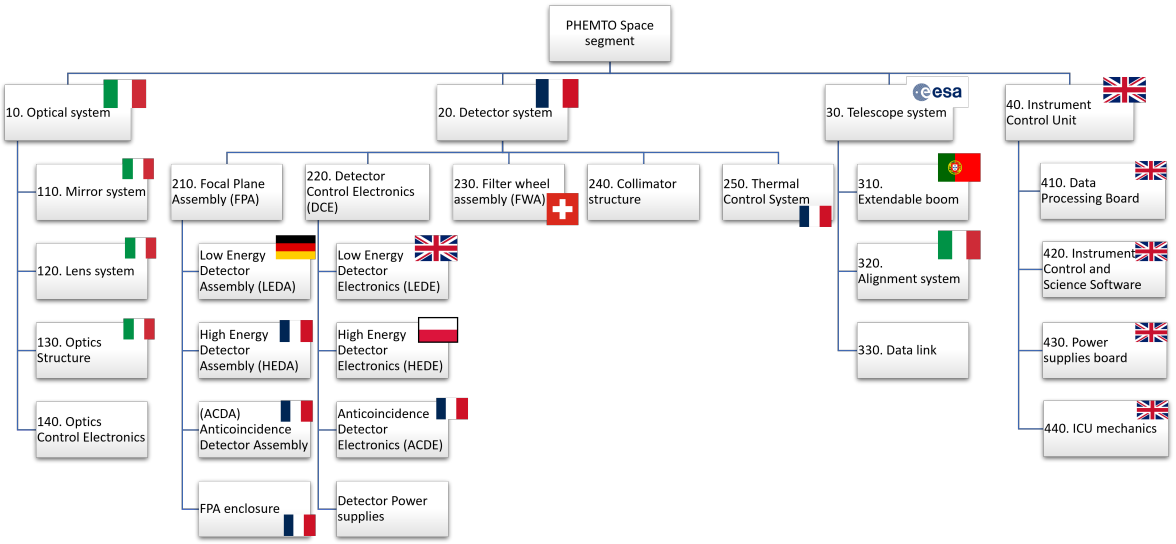


Fig. 6.1: PHEMTO space segment product tree with the different countries contribution flagged.

- Lens development has been pioneered by a few groups in Europe, including University of Ferrara (Italy) with contribution from INAF/OAS Bologna, which are now the European experts of this technology. They are responsible for the LARIX facility which is devoted to the Laue lens development and calibration.
- The Astrophysics Department of CEA (Commissariat à l'Energie Atomique; France), responsible for the HED development, has more than 60 years of experience in major space projects and is one of the major space laboratories in France, contributing to projects such as INTEGRAL, Solar Orbiter and many others.
- HLL, the SemiConductor Laboratory of the Max Planck Society is responsible for the LED development and has a European recognized expertize in developing silicon detectors since 30 years. It has provided semiconductor devices for many space missions such as XMM, BepiColombo, e-Rosita and is developing the detector of the WFI instrument onboard NewAthena.
- MSSL/UCL is responsible for the ICU. In the past 25 years, the electronics and software groups at MSSL have worked together to develop and build a number of leading-edge electronic control and data processing boxes for several space missions such as XMM, Swift, Solar Orbiter, Hinode. All of these instruments are still producing science data after 5-26 years of operation in space.
- The Space Research Center of the Polish Academy of Sciences (CBK PAN; Poland), responsible for the HEDE, has extensive experience in building space devices, as well as analyzing the data obtained. It has a long cooperation with CEA, providing the readout electronics of the CdTe pixel detectors and the instrument data processing for Solar Orbiter/STIX.
- The Université de Genève will provide the filter wheel assembly and has also a strong heritage in this domain as they designed, realized and delivered the Hitomi/SXS filter wheel and are developing filter wheels for the ESA THESEUS and NewAthena missions. University of Geneva has also a strong experience in Ground Segments, having created and managed the INTEGRAL Science Data Center during more than 25 years and organising the ground segment for several future mission like THESEUS.
- The Astroparticule and Cosmology Laboratory (APC, France), responsible for the ACD procurement, is a recently created laboratory in Paris, under the authorities of CNRS and Université Paris-Cité. Recently, the laboratory has conceived, developed and realized the $\gamma-$ray detector XGRE, onboard the French TARANIS satellite, launched in 2020.
- AST, the Active Space Technology company in Portugal will provide the extended boom. They

are also providing the 16 m extended boom for the ESA ENVISION/SRS instrument, which will launched in 2031.

7. Risk Assessment

No specific risk has been determined for the detector system as the parts constituting this system have already high heritage or follow a consolidated development plan through dedicated R&D programs. Also, the different teams contributing to this system have already a strong experience of working together. No risk also has been determined for the mirror which will be based upon similar procedures as the ones used for several space missions.

A risk may be linked to the complex mounting of the lens, that involves angular accuracies of a few arcseconds and positional tolerances at the level of tens of micrometers. As discussed in part 3.3.3, this risk is mitigated by on-going R&D studies in Italy. Another way to deal with this risk would be to diminish the PHEMTO focal length and thus release the requirements on this mounting. This will have the drawback of diminishing the PHEMTO energy range. Such option and its scientific impact will be studied in more details in the next phases.

Another envisaged risk is to have difficulties for the industry to provide a 20 m boom that satisfies the mechanical and thermal requirements. Here again, in agreement with the company, a mitigation will be to consider a smaller focal length, which will release the constraints on the boom.

8. Costing

Table 8.1: PHEMTO estimated ESA costs (in MEuros).

Launch vehicle (A62)	110
Spacecraft	280
Telescope system studies	30
Extendable boom	20
Operation	84
ESA project	72
Margin	72
Total CaC	668

Preliminary cost analysis of PHEMTO has ESA being responsible for the provision of the spacecraft, the launch vehicle, mission operations and the provision of ground stations. The spacecraft cost has been estimated using a conservative estimate of 200 keuros/kg for a dry mass of 1400 kg (see section 3.7). We have assumed an ESA cost of 30 Meuros for the telescope system studies and 20 Meuros for the extendable boom. No cost estimate was provided by AST as the cost will strongly depend on the boom mechanico-thermal constraints, which will be defined in Phase 0/A.

The ESA costs are given in table 8.1, with a total CaC of 668 Meuros, within the M-class mission cost ceiling of 670 Meuros. National agencies and PRODEX will fund the payload (detector + optics) provision and participation to the ground segment.

9. PHEMTO Core Members

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